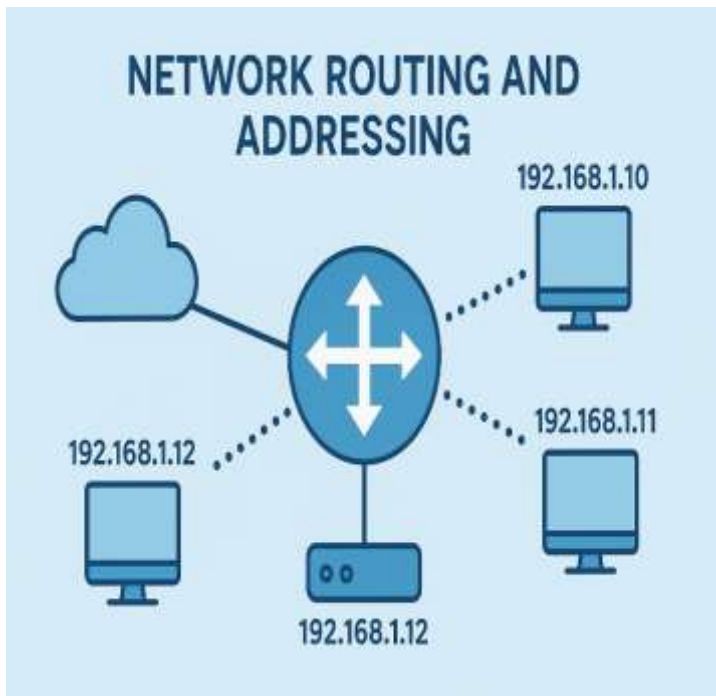




ROUTING AND ADDRESSING

Hands-on-Labs

ABSTRACT

This module focuses on networking in Google Cloud, specifically on routing, addressing, and traffic management. It covers advanced DNS configuration using geolocation-based traffic steering to optimize user experience and direct traffic efficiently. Additionally, it explores enabling secure private connectivity to Google services through Private Google Access and configuring Cloud NAT to allow outbound internet access for resources without public IP addresses. These labs provide practical experience in designing scalable, secure, and geographically-aware network architectures in Google Cloud.

DEBORAH BINYANYA

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CLOUD DNS - TRAFFIC STEERING USING GEOLOCATION POLICY

Overview

Cloud DNS routing policies enable users to configure DNS based traffic steering. A user can either create a **Weighted Round Robin (WRR)** routing policy or a **Geolocation (GEO) routing** policy. You can configure routing policies by creating special ResourceRecordSets with special routing policy values.

Use WRR to specify different weights per ResourceRecordSet for the resolution of domain names. Cloud DNS routing policies help ensure that traffic is distributed across multiple IP addresses by resolving DNS requests according to the configured weights.

In this lab, you will configure and test the Geolocation routing policy. Use GEO to specify source geolocations and to provide DNS answers corresponding to those geographies. The geolocation routing policy applies the nearest match for the source location when the traffic source location doesn't match any policy items exactly.

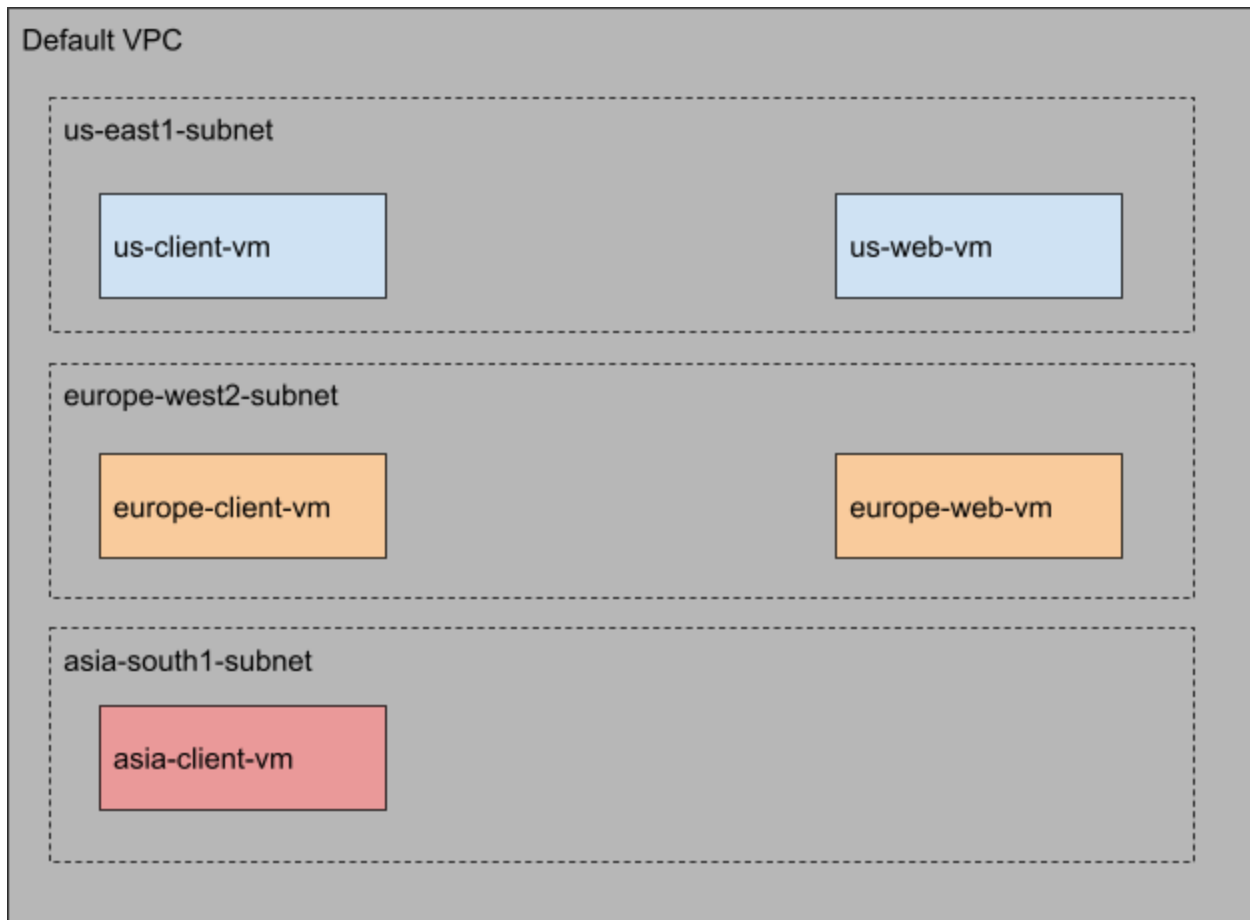
What you learn

You will learn how to:

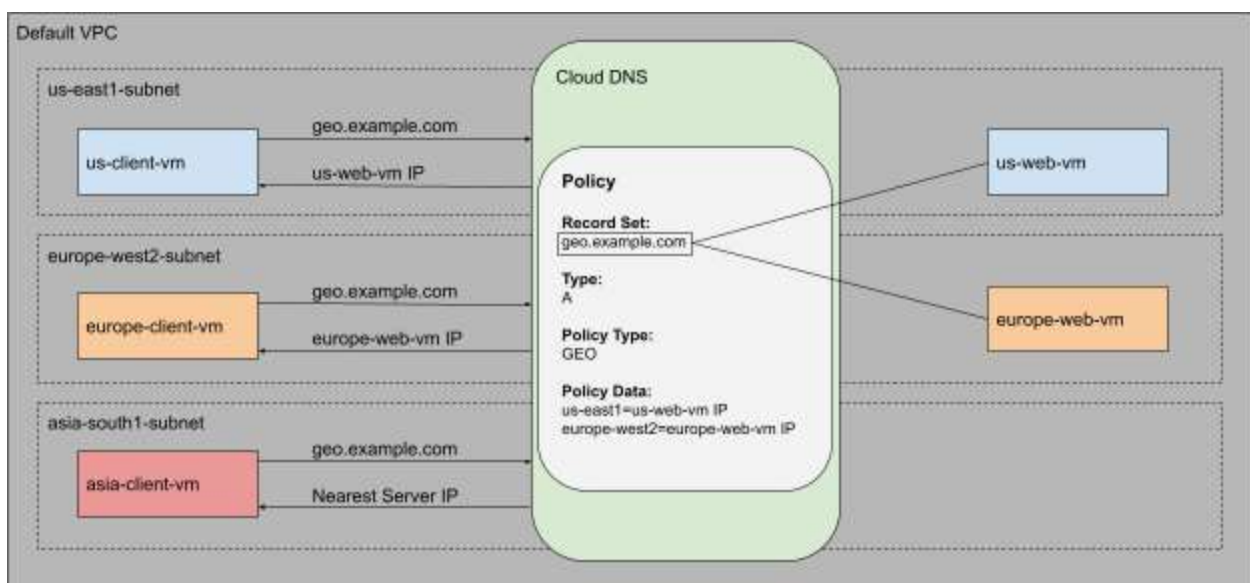
1. Launch client VMs, one in each region
2. Launch server VMs, one in each region except asia-south1
3. Create a private zone, for example.com
4. Create a Geolocation routing policy using gcloud commands
5. Test the configuration

Architecture

Use the default VPC network to create all the virtual machines (VM) and launch client VMs in 3 Google Cloud locations: one in the United States, another in Europe, and another in Asia. To demonstrate the Geolocation routing policy behavior, you will create the server VMs only in two of those location - in the United States and in Europe. The architecture will look similar to what is shown in the graphic. (Note that the actual regions and zones within the United States and Europe may differ from those shown in the graphic.)



You will use Cloud DNS routing policies and create ResourceRecordSets for `geo.example.com` and configure the Geolocation policy to help ensure that a client request is routed to a server in the client's closest region.



Setup and requirements

Before you click the Start Lab button

Note: Read these instructions.

Labs are timed and you cannot pause them. The timer, which starts when you click **Start Lab**, shows how long Google Cloud resources will be made available to you.

This Qwiklabs hands-on lab lets you do the lab activities yourself in a real cloud environment, not in a simulation or demo environment. It does so by giving you new, temporary credentials that you use to sign in and access Google Cloud for the duration of the lab.

What you need

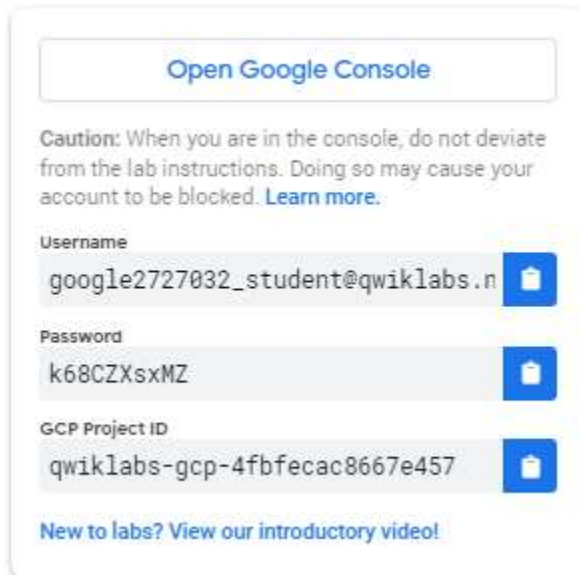
To complete this lab, you need:

- Access to a standard internet browser (Chrome browser recommended).
- Time to complete the lab.

Note: If you already have your own personal Google Cloud account or project, do not use it for this lab. **Note:** If you are using a Pixelbook, open an Incognito window to run this lab.

How to start your lab and sign in to the Console

1. Click the **Start Lab** button. If you need to pay for the lab, a pop-up opens for you to select your payment method. On the left is a panel populated with the temporary credentials that you must use for this lab.



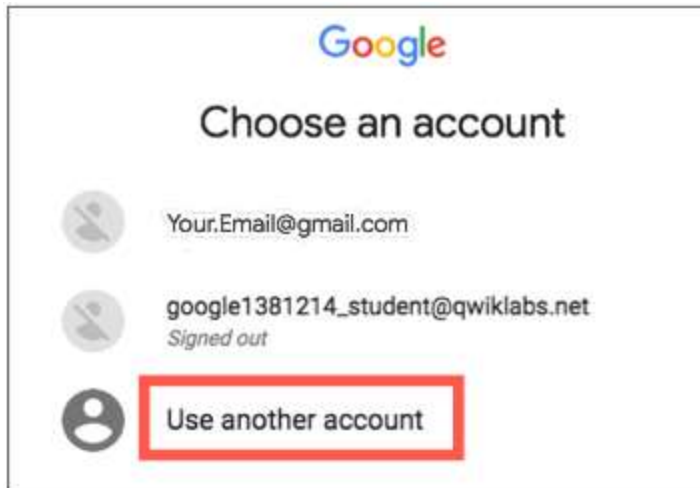
The screenshot shows a panel with the following content:

- A button at the top labeled "Open Google Console".
- A caution message: "Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)"
- A section for "Username" with the value "google2727032_student@qwiklabs.n" and a copy icon.
- A section for "Password" with the value "k68CZXsxMZ" and a copy icon.
- A section for "GCP Project ID" with the value "qwiklabs-gcp-4fbfecac8667e457" and a copy icon.
- A link at the bottom: "New to labs? View our introductory video!"

2. Copy the username, and then click **Open Google Console**. The lab spins up resources, and then opens another tab that shows the **Choose an account** page.

Note: Open the tabs in separate windows, side-by-side.

3. On the Choose an account page, click **Use Another Account**. The Sign in page opens.

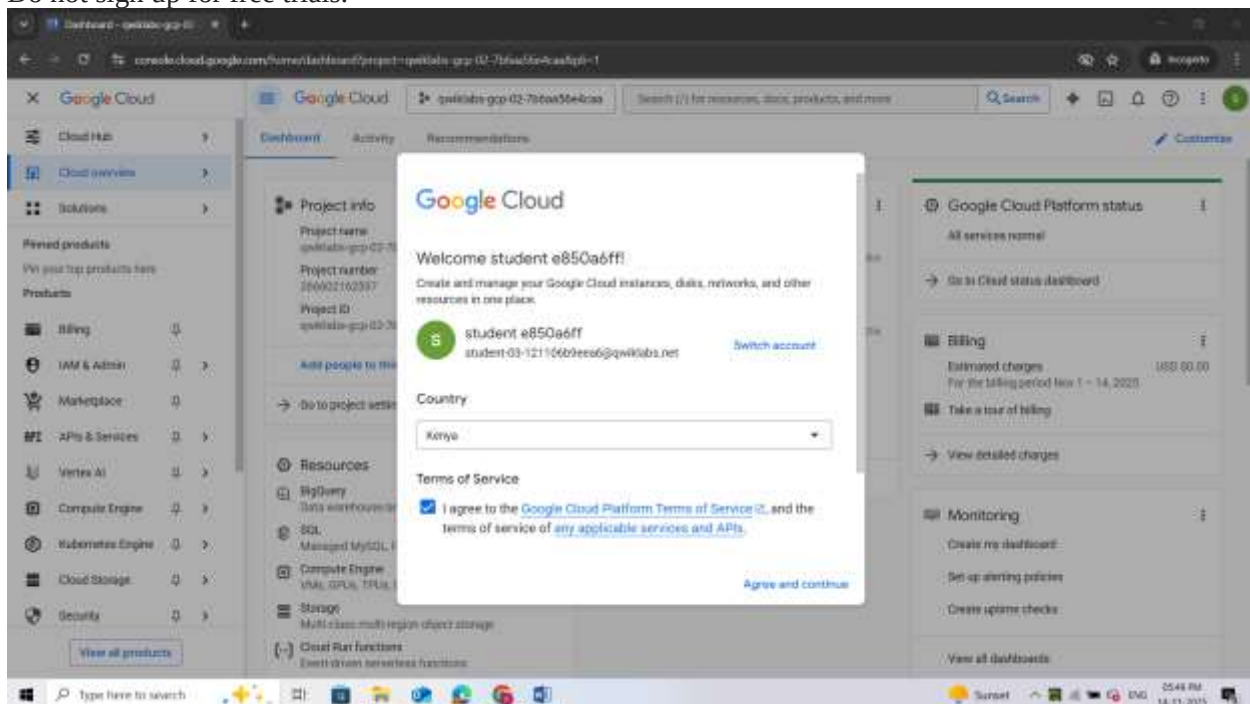


4. Paste the username that you copied from the Connection Details panel. Then copy and paste the password.

Note: You must use the credentials from the Connection Details panel. Do not use your Google Cloud Skills Boost credentials. If you have your own Google Cloud account, do not use it for this lab (avoids incurring charges).

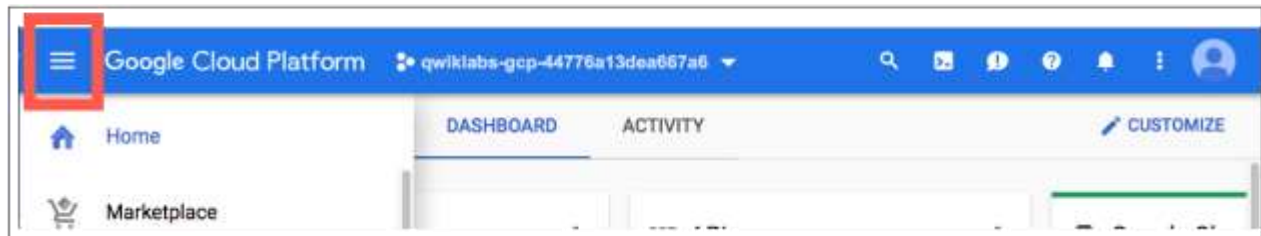
5. Click through the subsequent pages:

- Accept the terms and conditions.
- Do not add recovery options or two-factor authentication (because this is a temporary account).
- Do not sign up for free trials.



After a few moments, the Cloud console opens in this tab.

Note: You can view the menu with a list of Google Cloud Products and Services by clicking the **Navigation menu** at the top-left.

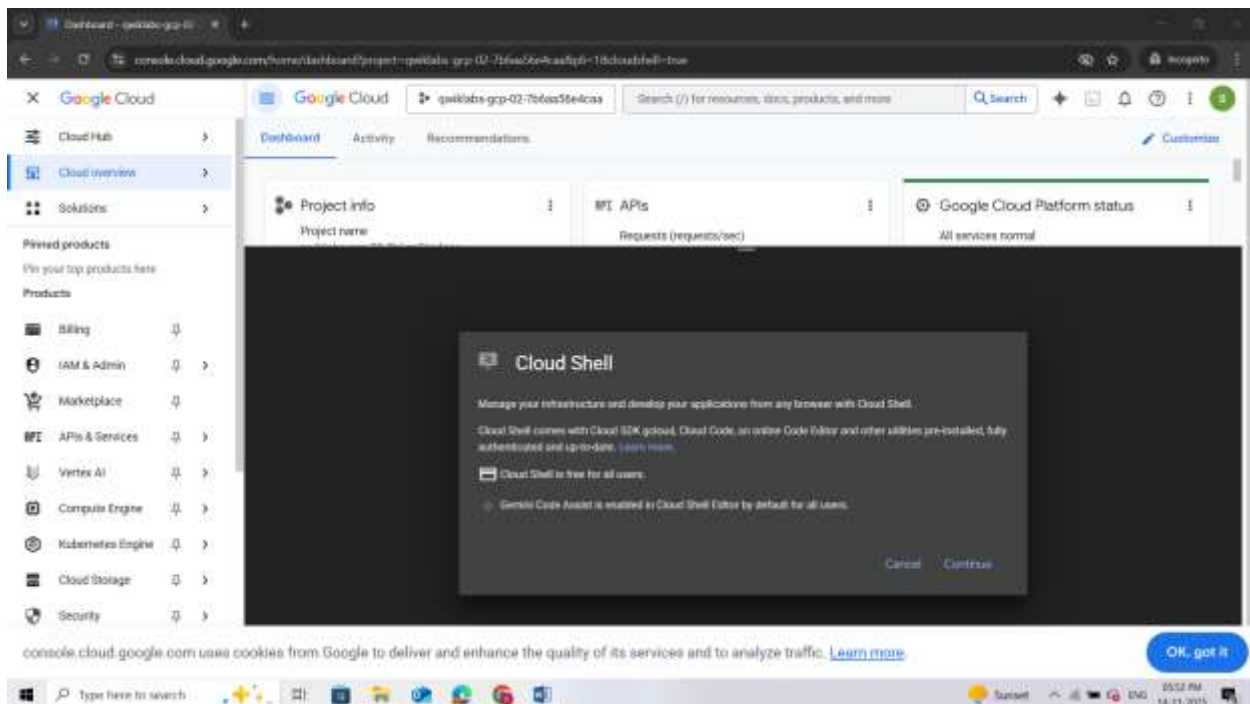


Activate Google Cloud Shell

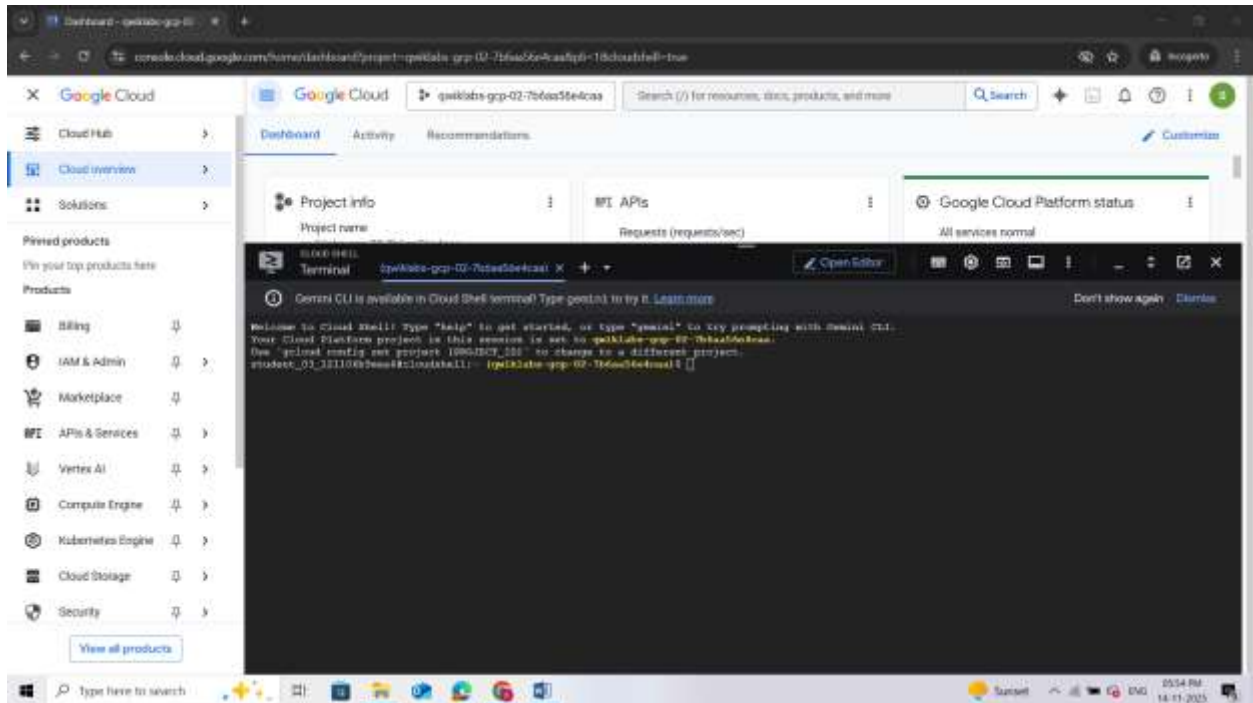
Google Cloud Shell is a virtual machine that is loaded with development tools. It offers a persistent 5GB home directory and runs on the Google Cloud.

Google Cloud Shell provides command-line access to your Google Cloud resources.

1. In Cloud console, on the top right toolbar, click the Open Cloud Shell button.
2. Click **Continue**.



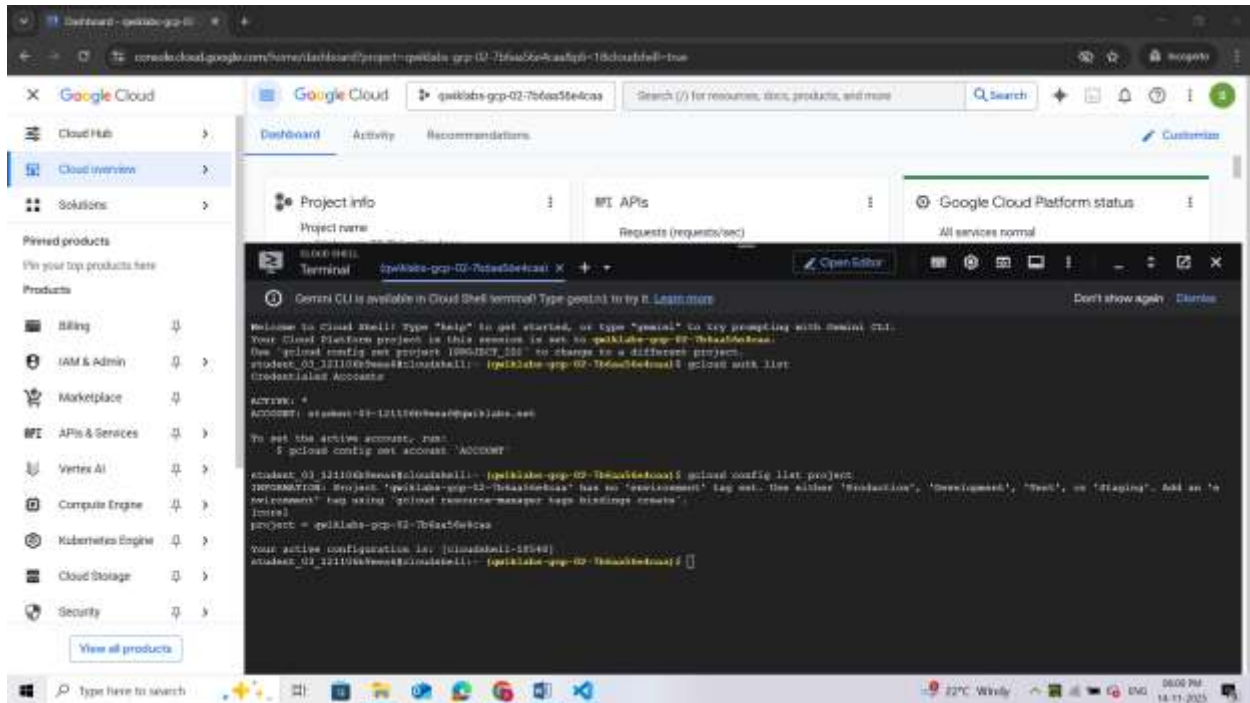
It takes a few moments to provision and connect to the environment. When you are connected, you are already authenticated, and the project is set to your *PROJECT_ID*. For example:



gcloud is the command-line tool for Google Cloud. It comes pre-installed on Cloud Shell and supports tab-completion.

- You can list the active account name with this command:
gcloud auth list
- You can list the project ID with this command:
gcloud config list project

Note: Full documentation of **gcloud** is available in the [gcloud CLI overview guide](#).



Task 1. Enable APIs

Ensure that the Compute and the Cloud DNS APIs are enabled. In this section, you will enable the APIs manually, using gcloud commands.

Enable Compute Engine API

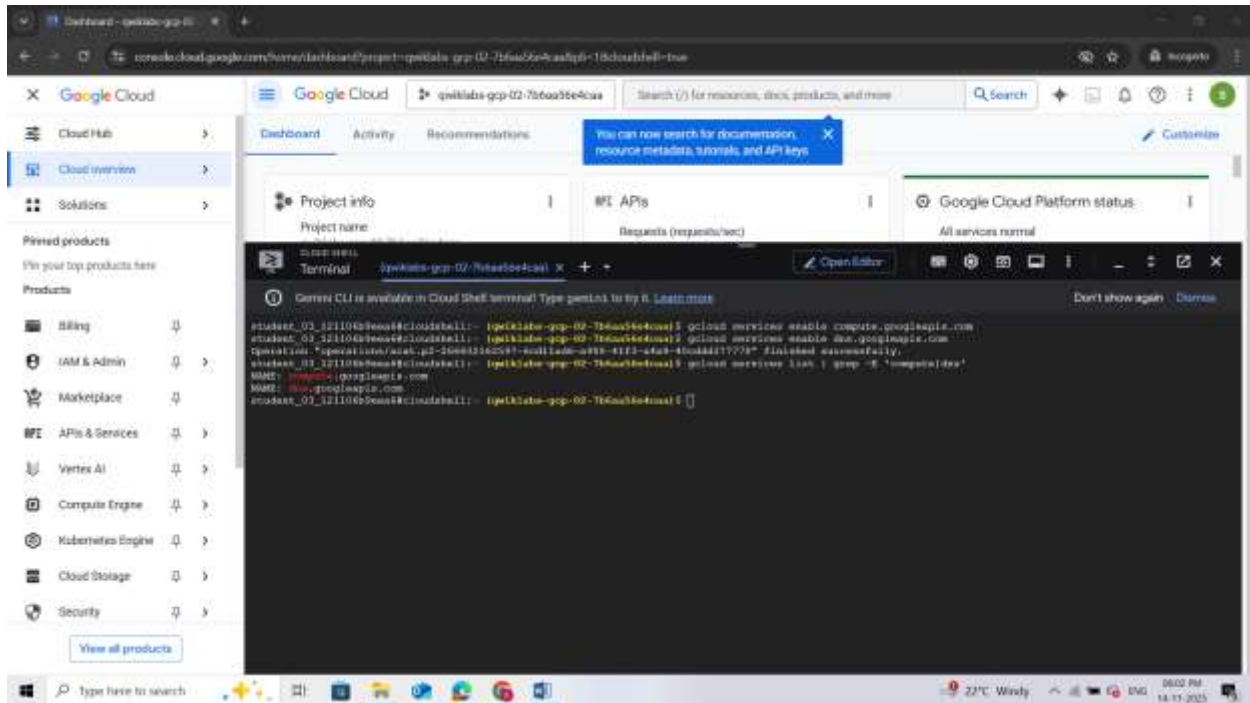
- Run the gcloud services enable command to enable the Compute Engine API:
`gcloud services enable compute.googleapis.com`
This command can take a few minutes to complete.

Enable Cloud DNS API

- Run the gcloud services enable command to enable the Cloud DNS API:
`gcloud services enable dns.googleapis.com`
This command can take a few minutes to complete.

Verify that the APIs are enabled

- Run the gcloud services list command to list all the enabled APIs. We should see compute.googleapis.com and dns.googleapis.com in the listed output.
`gcloud services list | grep -E 'compute|dns'`



Task 2. Configure the firewall

Before you create the client VMs and the web servers, you need to create two firewall rules.

Note: The firewall-rules create command can take a few minutes to complete. Please wait for the "Creating firewall...done" message before proceeding to the next step.

1. To be able to SSH into the client VMs, run the following to create a firewall rule to allow SSH traffic from Identity Aware Proxies (IAP):

```

gcloud compute firewall-rules create fw-default-iapproxy \
--direction=INGRESS \
--priority=1000 \
--network=default \
--action=ALLOW \
--rules=tcp:22,icmp \
--source-ranges=35.235.240.0/20

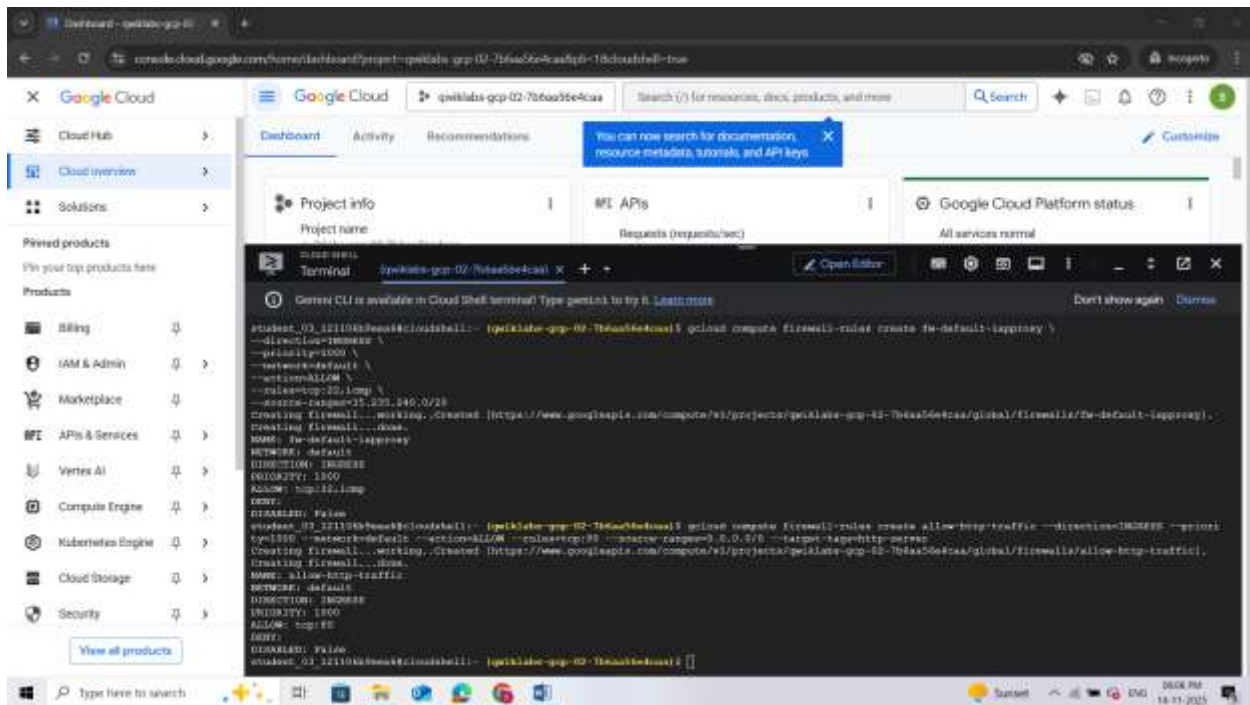
```

2. To allow HTTP traffic on the web servers, each web server will have a "http-server" tag associated with it. You will use this tag to apply the firewall rule only to your web servers:

```

gcloud compute firewall-rules create allow-http-traffic --direction=INGRESS --priority=1000 --
network=default --action=ALLOW --rules=tcp:80 --source-ranges=0.0.0.0/0 --target-tags=http-server

```



Task 3. Launch client VMs

Now that the APIs are enabled, and the firewall rules are in place, the next step is to set up the environment. In this section, you will create 3 client VMs, one in each region.

Launch a client in the United States

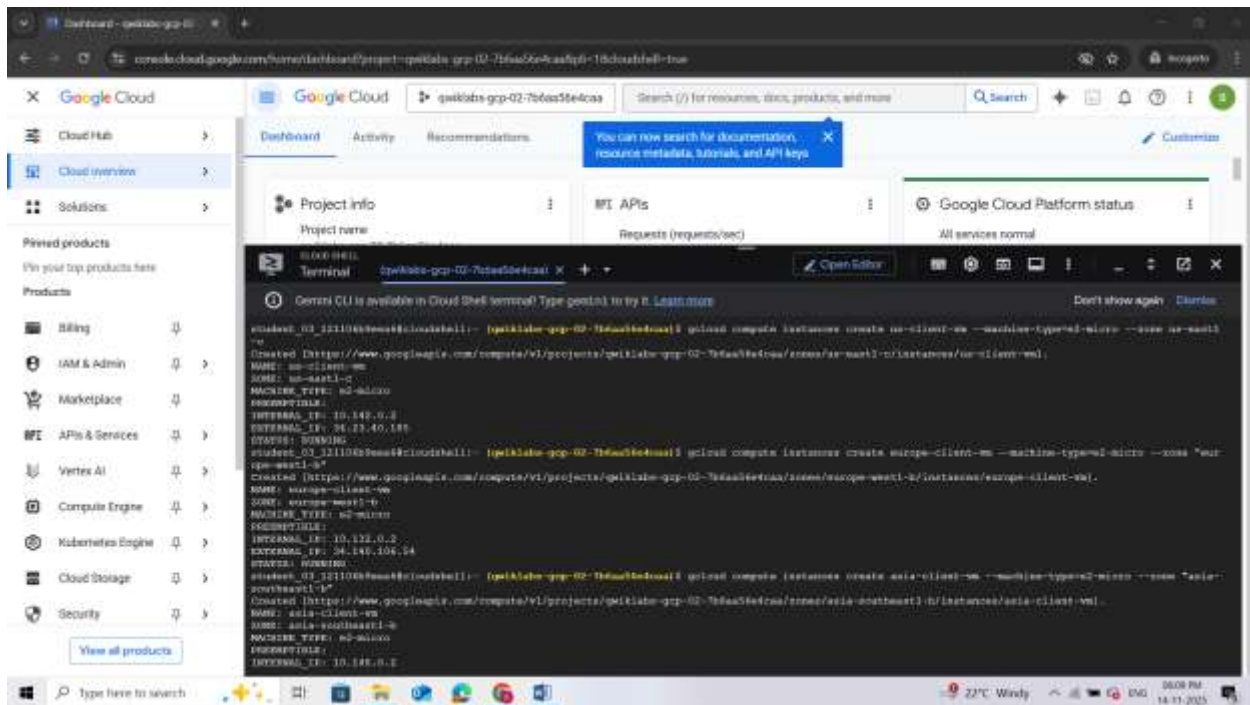
- Run the `gcloud compute instances create` command to create the client VMs:
`gcloud compute instances create us-client-vm --machine-type=e2-micro --zone "Zone 1"`
 This command can take a few minutes to complete. Please wait for a "Created" message before moving to the next step. Note that you may see a different zone in `gcloud` than in the sample output shown below.

Launch a client in Europe

- Run the following to create the client VMs:
`gcloud compute instances create europe-client-vm --machine-type=e2-micro --zone ""Zone 2""`
 Note that you may see a different zone in `gcloud` than in the sample output shown below.

Launch a client in Asia

- Run the following to create the client VMs:
`gcloud compute instances create asia-client-vm --machine-type=e2-micro --zone ""Zone 3""`
 Note that you may see a different zone in `gcloud` than in the sample output shown below.



Task 4. Launch Server VMs

Now that the client VM's are up and running, the next step is to create the server VMs. You will use a startup script to configure and set up the web servers. As mentioned earlier, you will create the server VMs only in 2 regions: us-east1 and europe-west2.

- Run the gcloud compute instances create command to create the server VMs. The compute instance create command can take a few minutes to complete. Please wait for a "Created" message before moving to the next step.

Launch server in the United States

- Run the following command:

```
gcloud compute instances create us-web-vm \
--machine-type=e2-micro \
--zone="Zone 1" \
--network=default \
--subnet=default \
--tags=http-server \
--metadata=startup-script='#!/bin/bash
apt-get update
apt-get install apache2 -y
echo "Page served from: "Region 1"" | \
tee /var/www/html/index.html
systemctl restart apache2'
```

Note that you may see a different zone in gcloud than in the sample output shown below.

Launch server in Europe

- Run the following to command:
`gcloud compute instances create europe-web-vm \`
`--machine-type=e2-micro \`
`--zone="Zone 2" \`
`--network=default \`
`--subnet=default \`
`--tags=http-server \`
`--metadata=startup-script='#!/bin/bash`
`apt-get update`
`apt-get install apache2 -y`
`echo "Page served from: "Zone 2"" | \`
`tee /var/www/html/index.html`
`systemctl restart apache2'`

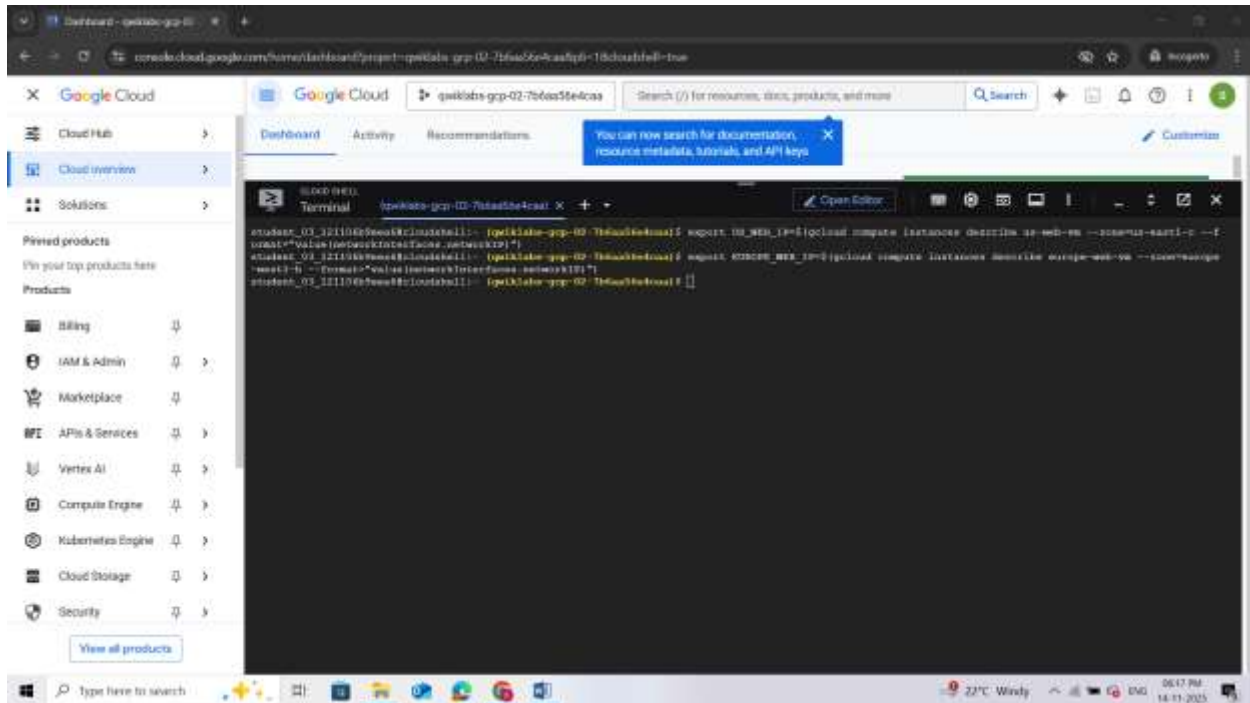
Note that you may see a different zone in gcloud than in the sample output shown below.

The screenshot shows a Google Cloud Shell terminal window. The terminal displays the command to create a VM instance in Europe: `gcloud compute instances create europe-web-vm --machine-type=e2-micro --zone="Zone 2" --network=default --subnet=default --tags=http-server --metadata=startup-script='#!/bin/bash apt-get update apt-get install apache2 -y echo "Page served from: "Zone 2"" | tee /var/www/html/index.html systemctl restart apache2'`. The output shows the instance is being created in the `us-east1-c` zone, which is different from the specified `Zone 2`. The instance details shown are: `NAME: us-east1-c`, `MACHINE_TYPE: e2-micro`, `INTERNAL_IP: 10.142.0.3`, and `EXTERNAL_IP: 35.245.211.44`. The status is `STATUS: RUNNING`.

Task 5. Setting up environment variables

Before you configure Cloud DNS, note the Internal IP addresses of the web servers. You need these IPs to create the routing policy. In this section, you will use the `gcloud compute instances describe` command to save the internal IP addresses as environment variables.

- Command to save IP address for the VM in the United States
`export US_WEB_IP=$(gcloud compute instances describe us-web-vm --zone="Zone 1" --format="value(networkInterfaces.networkIP)")`
- Command to save the IP address for the VM in Europe:
`export EUROPE_WEB_IP=$(gcloud compute instances describe europe-web-vm --zone="Zone 2" --format="value(networkInterfaces.networkIP)")`



Task 6. Create the private zone

Now that your client and server VMs are running, it's time to configure the DNS settings. Before creating the A records for the web servers, you need to create the Cloud DNS Private Zone.

For this lab, use the example.com domain name for the Cloud DNS zone.

- Use the `gcloud dns managed-zones create` command to create the zone:
`gcloud dns managed-zones create example --description=test --dns-name=example.com --networks=default --visibility=private`



Task 7. Create Cloud DNS Routing Policy

In this section, configure the Cloud DNS Geolocation Routing Policy. You will create a record set in the example.com zone that you created in the previous section.

Create

- Use the `gcloud dns record-sets create` command to create the geo.example.com recordset:
`gcloud dns record-sets create geo.example.com \`
`--ttl=5 --type=A --zone=example \`
`--routing-policy-type=GEO \`
`--routing-policy-data=""Region 1"=$US_WEB_IP;"Region 2"=$EUROPE_WEB_IP"`

You are creating an A record with a Time to Live (TTL) of 5 seconds. The policy type is GEO, and the `routing_policy_data` field accepts a semicolon-delimited list of the format `${region}:${rrdata},${rrdata}`.

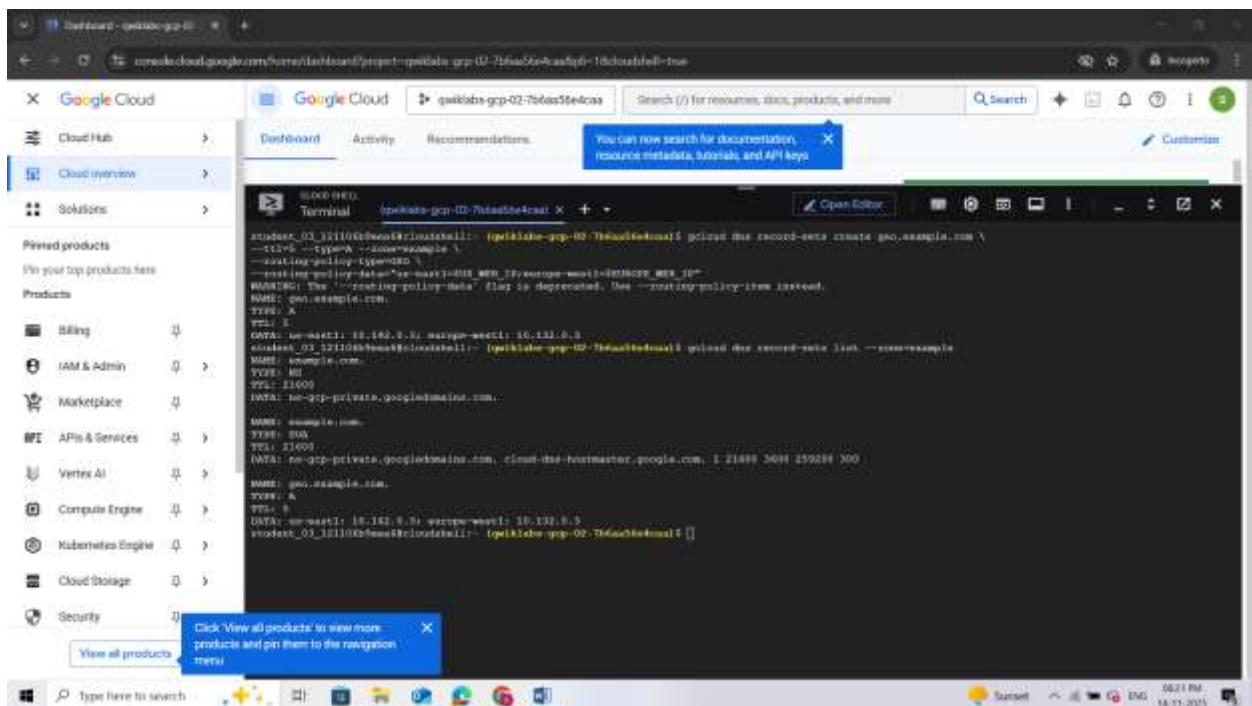
Verify

- Use the `dns record-sets list` command to verify that the `geo.example.com` DNS record is configured as expected:

`gcloud dns record-sets list --zone=example`

The output shows that an A record with a TTL of 5 is created for `geo.example.com`, and the data matches our server set up in each region.

Note that in `gcloud`, the `DATA` value under `geo.example.com` may include United States and Europe regions that differ from the sample output below.



```
student_01_1211045@cloudshell:~$ gcloud dns record-sets list --zone=example \
--ttl=5 --type=A --zone=example \
--routing-policy-type=RR \
--routing-policy-data="us-east1-dns-key:10.142.0.1;eu-west1-dns-key:10.132.0.1"
WARNING: The "--routing-policy-data" flag is deprecated. Use "--routing-policy-item" instead.
NAME: geo.example.com.
TYPE: A
TTL: 5
DATA: us-east1: 10.142.0.1; eu-west1: 10.132.0.1
student_01_1211045@cloudshell:~$ gcloud dns record-sets list --zone=example
NAME: geo.example.com.
TYPE: RR
TTL: 11000
DATA: us-gcp-private.googleusercontent.com.
NAME: geo.example.com.
TYPE: A
TTL: 5
DATA: us-east1: 10.142.0.1; eu-west1: 10.132.0.1
student_01_1211045@cloudshell:~$
```

Task 8. Testing

It's time to test the configuration. In this section, you will SSH into all the client VMs. Since all of the web server VMs are behind the `geo.example.com` domain, you will use `CURL` command to access this endpoint.

Since you are using a Geolocation policy, the expected result is that:

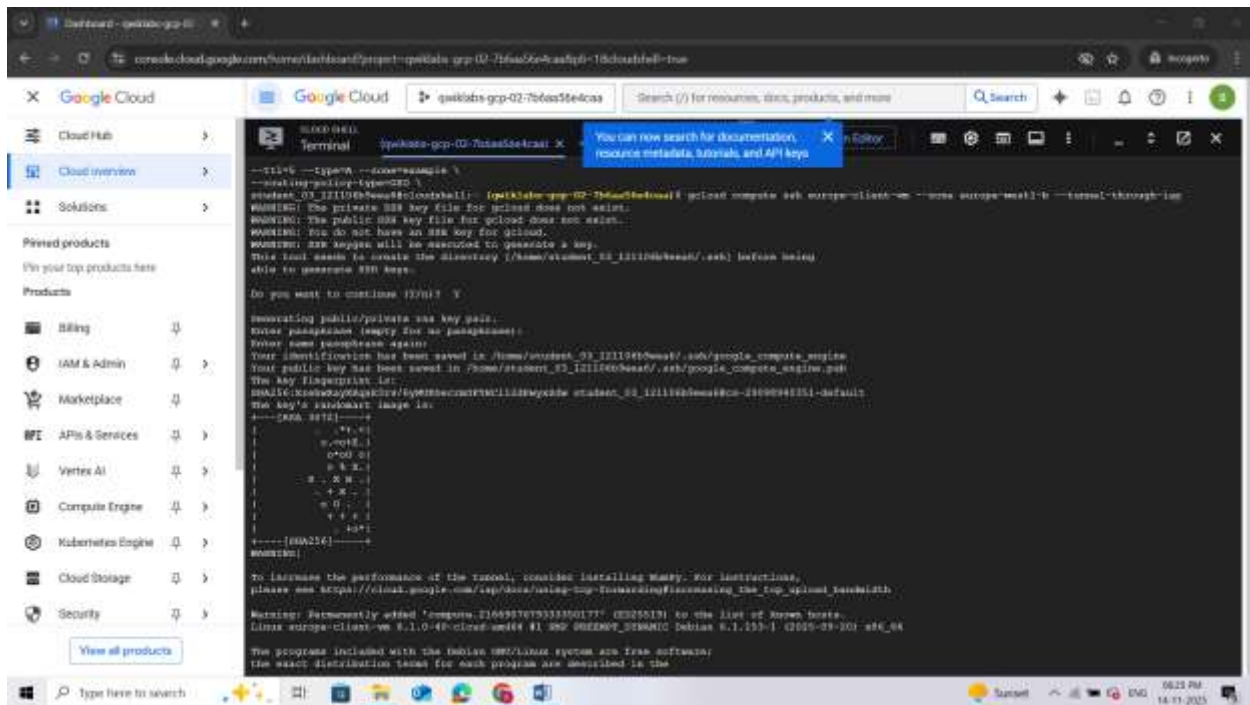
- The client in the US should always get a response from the `<Zone 1` region.
- The client in Europe should always get a response from the `Zone 2` region.

Testing from the client VM in Europe

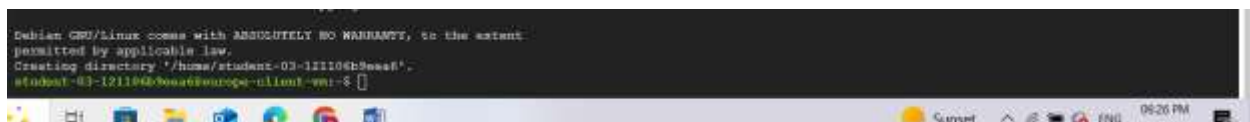
- Use the `gcloud compute ssh` command to log into the client VM:

`gcloud compute ssh europe-client-vm --zone "Zone 2" --tunnel-through-iap`

- Follow prompts to SSH into the machine. When asked to enter the passphrase, leave the field blank and press the Enter key twice.



Once complete, the command line should change to "user_name@europe-client-vm:~\$"

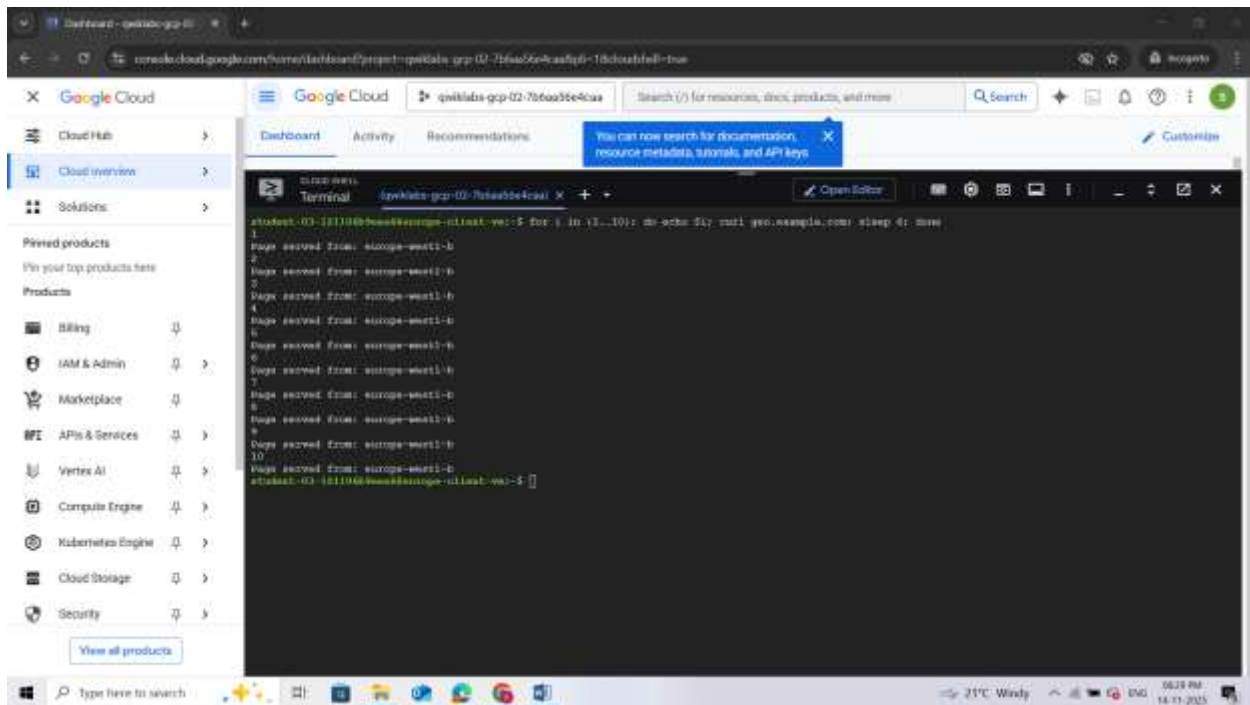


Use curl to access the web server

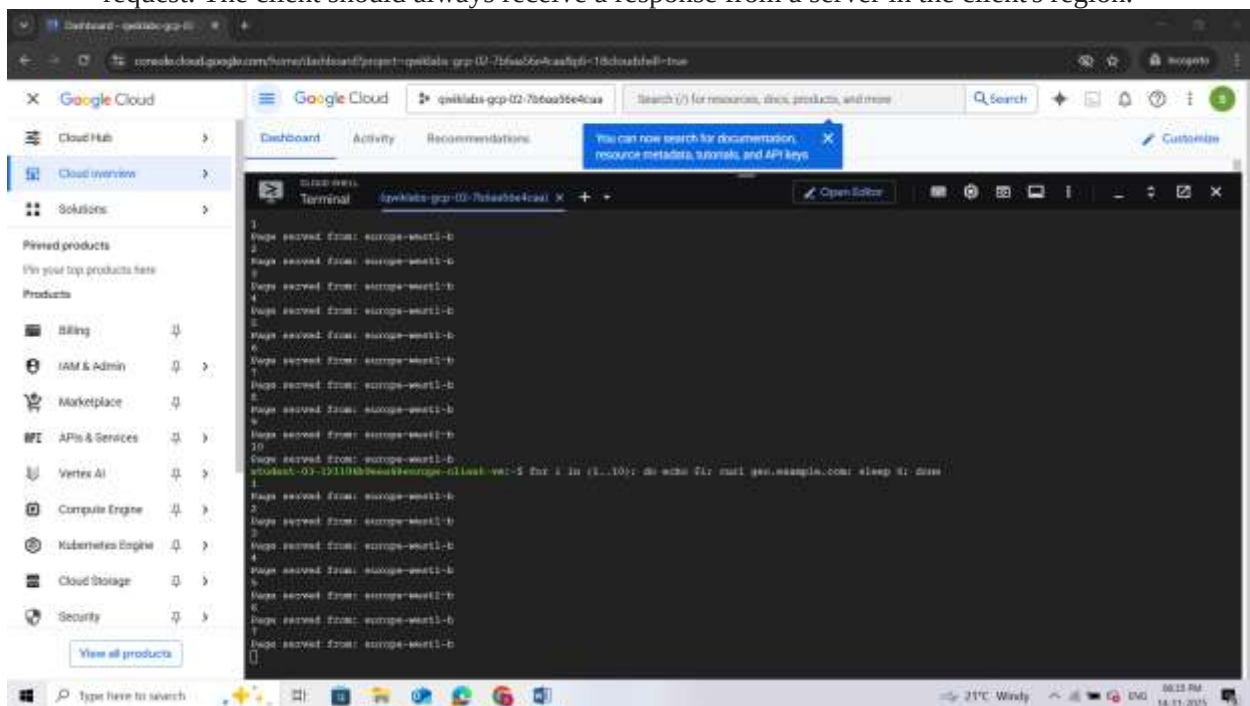
- Now that you are in the client VM, use the CURL command to access the geo.example.com endpoint. The loop is configured to run the command ten times with a sleep timer of 6 seconds:

for i in {1..10}; do echo \$i; curl geo.example.com; sleep 6; done

Since the TTL on the DNS record is set to 5 seconds, a sleep timer of 6 seconds has been added. The sleep timer will make sure that you get an uncached DNS response for each cURL request. This command will take approximately one minute to complete.



2. Run this test multiple times and analyze the output to see which server is responding to the request. The client should always receive a response from a server in the client's region.



Getting back to Cloud Shell

- Once you have run the test multiple times, exit the client VM in Europe by typing "exit" in the VM's command prompt. This will bring you back to the Cloud Shell console.



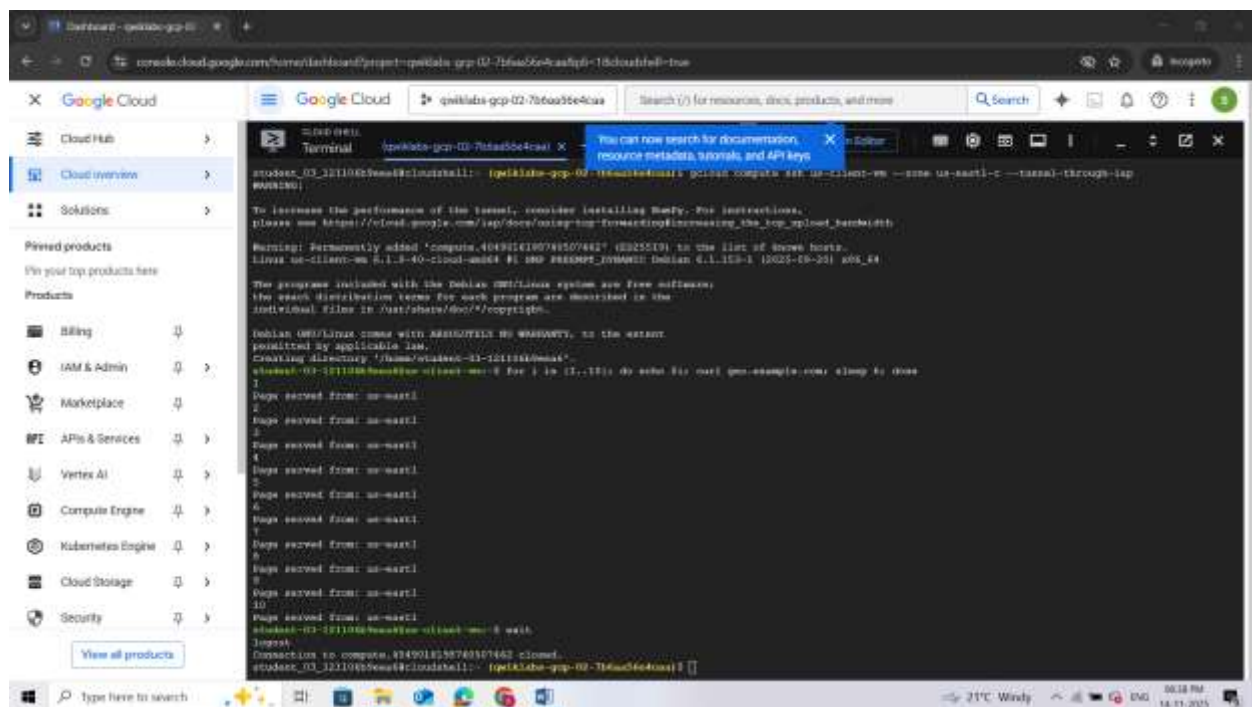
```
CLOUD SHELL
Terminal (qwiklabs-gcp-02-7b6aa56e4caa) x + v Open Editor

student-03-121106b9eea6@europa-client-vm:~$ exit
logout
Connection to compute.2166907679333350177 closed.
student_03_121106b9eea6@cloudshell:~ (qwiklabs-gcp-02-7b6aa56e4caa) $
```

Testing from the client VM in us-east1

Now perform the same test from the client VM in the US.

1. Use the gcloud command below to SSH into the us-client-vm:
`gcloud compute ssh us-client-vm --zone "Zone 1" --tunnel-through-iap`
2. Use the curl command to access geo.example.com:
`for i in {1..10}; do echo $i; curl geo.example.com; sleep 6; done`
3. Now analyze the output to see which server is responding to the request. The client should always receive a response from a server in the client's region. The expected output is "Page served from: <filled at lab start>".
4. Once you have run the test multiple times, exit the client VM in the US by typing "exit" in the VM's command prompt.



```
Google Cloud
Cloud Hub
Cloud overview
Solutions
Pinned products
View your top products here
Products
Billing
IAM & Admin
Marketplace
APIs & Services
Vertex AI
Compute Engine
Kubernetes Engine
Cloud Storage
Security

Terminal (qwiklabs-gcp-02-7b6aa56e4caa) x + v Open Editor

student-03-121106b9eea6@cloudshell:~ (qwiklabs-gcp-02-7b6aa56e4caa) $ gcloud compute ssh us-client-vm --zone us-east1-c --tunnel-through-iap
WARNING: Permanently added 'compute.404918158740507442' (2025519) to the list of known hosts.
Linux us-client-vm 4.15.0-40-cloud-amd64 #1 SMP PREEMPT_DYNAMIC x86_64 GNU/Linux

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/*-copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Creating directory /home/student-03-121106b9eea6.
student-03-121106b9eea6@us-client-vm:~$ for i in {1..10}; do echo $i; curl geo.example.com; sleep 6; done
1
Page served from: us-east1
2
Page served from: us-east1
3
Page served from: us-east1
4
Page served from: us-east1
5
Page served from: us-east1
6
Page served from: us-east1
7
Page served from: us-east1
8
Page served from: us-east1
9
Page served from: us-east1
10
Page served from: us-east1
student-03-121106b9eea6@us-client-vm:~$ exit
logout
Connection to compute.404918158740507442 closed.
student_03_121106b9eea6@cloudshell:~ (qwiklabs-gcp-02-7b6aa56e4caa) $
```

Testing from the client VM in Asia

So far you have tested the setup from the United States and Europe. You have servers running in both the regions and have matching record sets for both the regions in Cloud DNS routing policy. There is no matching policy item for the region within Asia (selected earlier) in the Cloud DNS routing policy.

The Geolocation policy will apply a "nearest" match for source location when the source of the traffic doesn't match any policy items exactly. This means that the Asia client should be directed to the nearest web server.

In this section, you will resolve the geo.example.com domain from the client VM in Asia and will analyze the response.

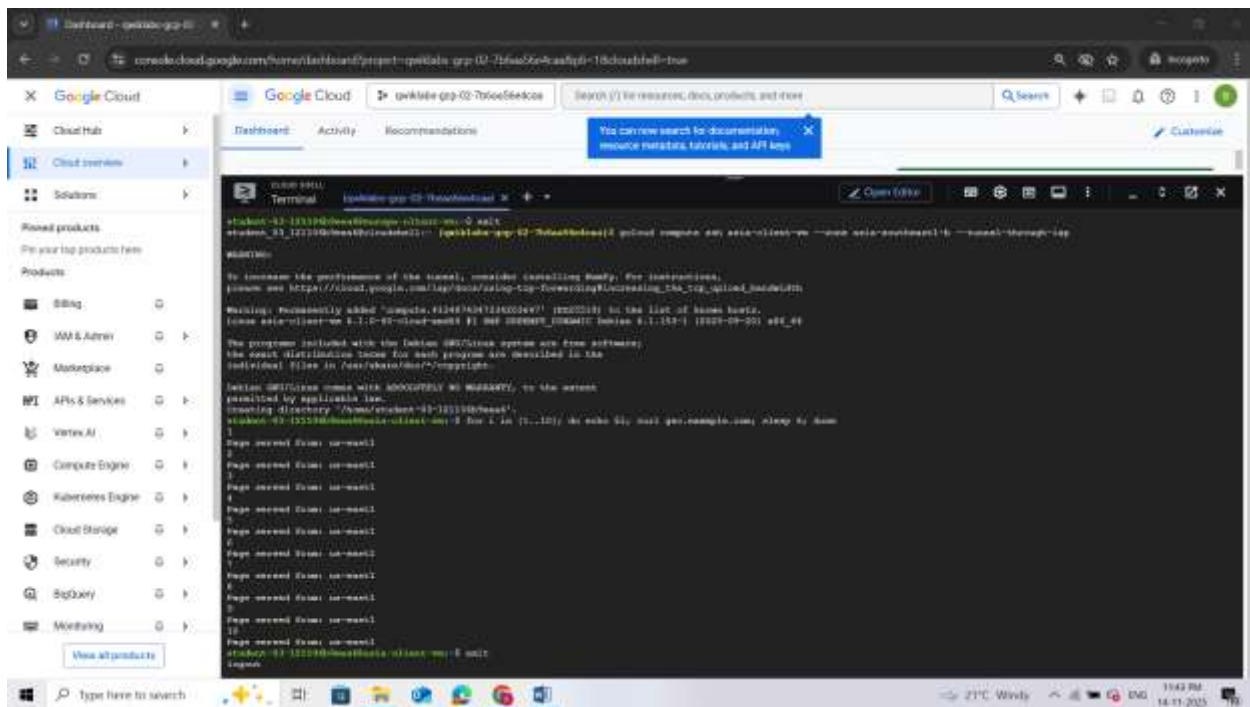
1. SSH into the asia-client-vm. For <SELECTED-ZONE>, use the zone that you used to create the Asia client.

`gcloud compute ssh asia-client-vm --zone <SELECTED-ZONE> --tunnel-through-iap`

2. Then access geo.example.com:

`for i in {1..10}; do echo $i; curl geo.example.com; sleep 6; done`

3. Analyze the output to see which server is responding to the request. Since there is no policy item for any of the Asia regions, Cloud DNS will direct the client to the nearest server.



4. Once you have run the test multiple times, exit the client VM in Asia by typing "exit" in the VM's command prompt.

```
Page solved from: 40.128.128.128
student-03-121106b9eea6@asia-client-vm:~$ exit
logout
Connection to compute.4134874347334203647 closed.
```

Task 9. Delete lab resources

Although all resources you used in this lab will be deleted when you finish, it is good practice to remove resources you no longer need to avoid unnecessary charges.

- The following gcloud commands will delete all the resources that were created in the lab. (Note that SELECTED-ZONE is the Asia zone that you wrote down earlier.)

#delete VMS

`gcloud compute instances delete -q us-client-vm --zone "ZONE"`

`gcloud compute instances delete -q us-web-vm --zone "ZONE"`

`gcloud compute instances delete -q europe-client-vm --zone "Zone 2"`

`gcloud compute instances delete -q europe-web-vm --zone "Zone 2"`

`gcloud compute instances delete -q asia-client-vm --zone SELECTED-ZONE`

#delete FW rules

`gcloud compute firewall-rules delete -q allow-http-traffic`

`gcloud compute firewall-rules delete fw-default-iapproxy`

#delete record set

`gcloud dns record-sets delete geo.example.com --type=A --zone=example`

#delete private zone

`gcloud dns managed-zones delete example`

```

student_01_12234@aws01-100000000000: ~$ gcloud compute instances delete -q us-client-vm --zone "ZONE"
Deleted https://www.googleapis.com/compute/v1/projects/gcp-lab-us-central1/instances/us-client-vm.
The following instance will be deleted:
  - [us-client-vm]

In the next 60 seconds, the instance will be deleted.

student_01_12234@aws01-100000000000: ~$ gcloud compute instances delete -q us-web-vm --zone "ZONE"
Deleted https://www.googleapis.com/compute/v1/projects/gcp-lab-us-central1/instances/us-web-vm.
The following instance will be deleted:
  - [us-web-vm]

In the next 60 seconds, the instance will be deleted.

student_01_12234@aws01-100000000000: ~$ gcloud compute instances delete -q europe-client-vm --zone "Zone 2"
Deleted https://www.googleapis.com/compute/v1/projects/gcp-lab-europe-west1/instances/europe-client-vm.
The following instance will be deleted:
  - [europe-client-vm]

In the next 60 seconds, the instance will be deleted.

student_01_12234@aws01-100000000000: ~$ gcloud compute instances delete -q europe-web-vm --zone "Zone 2"
Deleted https://www.googleapis.com/compute/v1/projects/gcp-lab-europe-west1/instances/europe-web-vm.
The following instance will be deleted:
  - [europe-web-vm]

In the next 60 seconds, the instance will be deleted.

student_01_12234@aws01-100000000000: ~$ gcloud compute instances delete -q asia-client-vm --zone SELECTED-ZONE
Deleted https://www.googleapis.com/compute/v1/projects/gcp-lab-asia-south1/instances/asia-client-vm.
The following instance will be deleted:
  - [asia-client-vm]

In the next 60 seconds, the instance will be deleted.

student_01_12234@aws01-100000000000: ~$ gcloud compute firewall-rules delete -q allow-http-traffic
Deleted https://www.googleapis.com/compute/v1/projects/gcp-lab-us-central1/firewalls/allow-http-traffic.
The following firewall rule will be deleted:
  - [allow-http-traffic]

In the next 60 seconds, the firewall rule will be deleted.

student_01_12234@aws01-100000000000: ~$ gcloud compute firewall-rules delete fw-default-iapproxy
Deleted https://www.googleapis.com/compute/v1/projects/gcp-lab-us-central1/firewalls/fw-default-iapproxy.
The following firewall rule will be deleted:
  - [fw-default-iapproxy]

In the next 60 seconds, the firewall rule will be deleted.

student_01_12234@aws01-100000000000: ~$ gcloud dns record-sets delete geo.example.com --type=A --zone=example
Deleted https://www.googleapis.com/dns/v1/projects/gcp-lab-us-central1/zones/example.com/recordsets/geo.example.com/A.
The following record set will be deleted:
  - [geo.example.com/A]

In the next 60 seconds, the record set will be deleted.

student_01_12234@aws01-100000000000: ~$ gcloud dns managed-zones delete example
Deleted https://www.googleapis.com/dns/v1/projects/gcp-lab-us-central1/managedZones/example.
The following managed zone will be deleted:
  - [example]

In the next 60 seconds, the managed zone will be deleted.

```

Review

In this lab, you configured and used Cloud DNS routing policies with Geolocation routing policy. You also verified the configuration and behavior of the Cloud DNS routing policy by observing the HTTP response when accessing the web servers.

[Next steps / Learn more](#)

You can read more about [Cloud DNS routing policies](#).

[End your lab](#)

When you have completed your lab, click **End Lab**. Google Cloud Skills Boost removes the resources you've used and cleans the account for you.

QUIZ

- ✓ 1. You must create a VM that has an IPv6 address. How do you do it?
- ✓ ☒ Create a dual-stack subnet, and create the VM with an IPv6 address.
 - ☐ Create a single-stack network, and create the VM with an IPv6 address.
 - ☐ Create an IPv6-only subnet, and create the VM with an IPv6 address.
 - ☐ Create a dual-stack network, and create the VM with an IPv6 address.

This is correct. Dual-stack subnets support both IPv4 and IPv6, allowing you to create VMs with both types of addresses.

- ✓ 2. To set up hybrid deployments for DNS resolution, which type of DNS policy should you use?
- ☐ Response policy
 - ☐ Routing policy
 - ☐ Traffic policy
 - ✓ ☒ Server policy

This is correct. Server policies are used to define which DNS servers should be queried for different DNS zones, making them suitable for hybrid deployments where you might have on-premises and cloud-based DNS servers.

LAB 2: Implement Private Google Access and Cloud NAT

Overview

In this lab, you implement Private Google Access and Cloud NAT for a VM instance that doesn't have an external IP address. Then, you verify access to public IP addresses of Google APIs and services and other connections to the internet.

VM instances without external IP addresses are isolated from external networks. Using Cloud NAT, these instances can access the internet for updates and patches, and in some cases, for **bootstrapping**. As a managed service, Cloud NAT provides high availability without user management and intervention.

Objectives

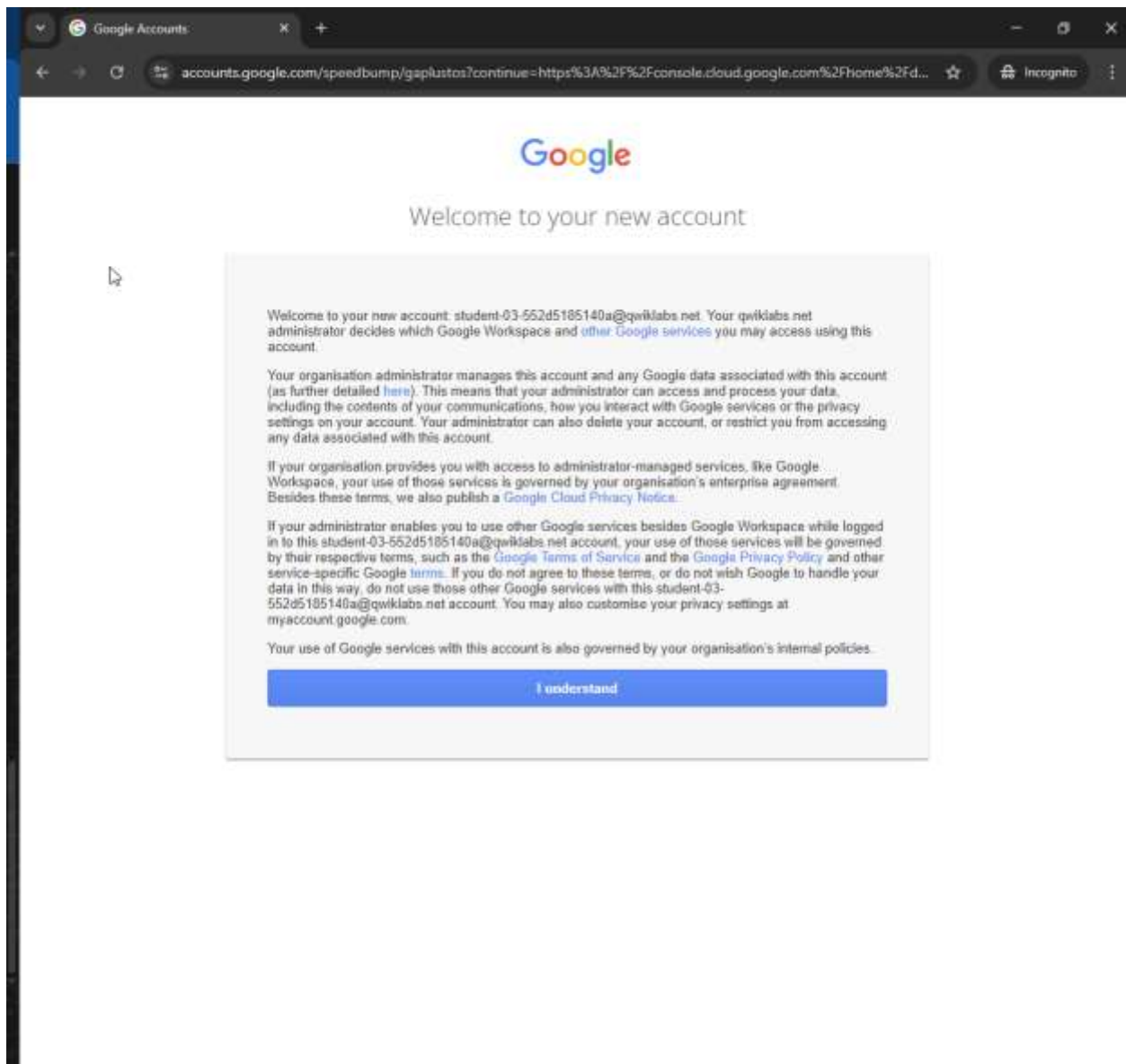
In this lab, you learn how to perform the following tasks:

- Configure a VM instance that doesn't have an external IP address
- Connect to a VM instance using an Identity-Aware Proxy (IAP) tunnel
- Enable Private Google Access on a subnet
- Configure a Cloud NAT gateway
- Verify access to public IP addresses of Google APIs and services and other connections to the internet

Setup and Requirements

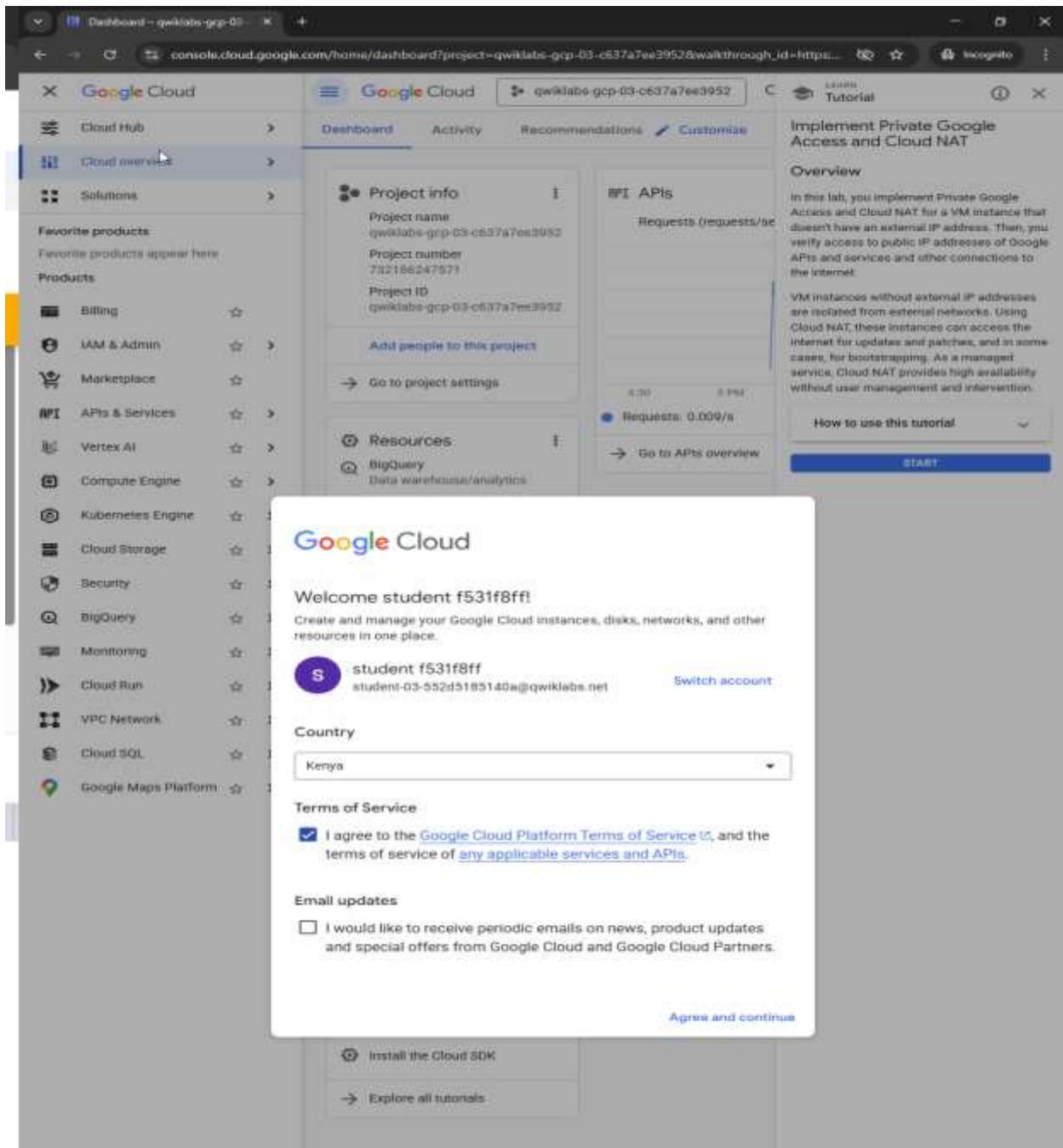
For each lab, you get a new Google Cloud project and set of resources for a fixed time at no cost.

1. Click the **Start Lab** button. If you need to pay for the lab, a pop-up opens for you to select your payment method. On the left is the **Lab Details** panel with the following:
 - The **Open Google Cloud console** button
 - Time remaining
 - The temporary credentials that you must use for this lab
 - Other information, if needed, to step through this lab
2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window** if you are running the Chrome browser).



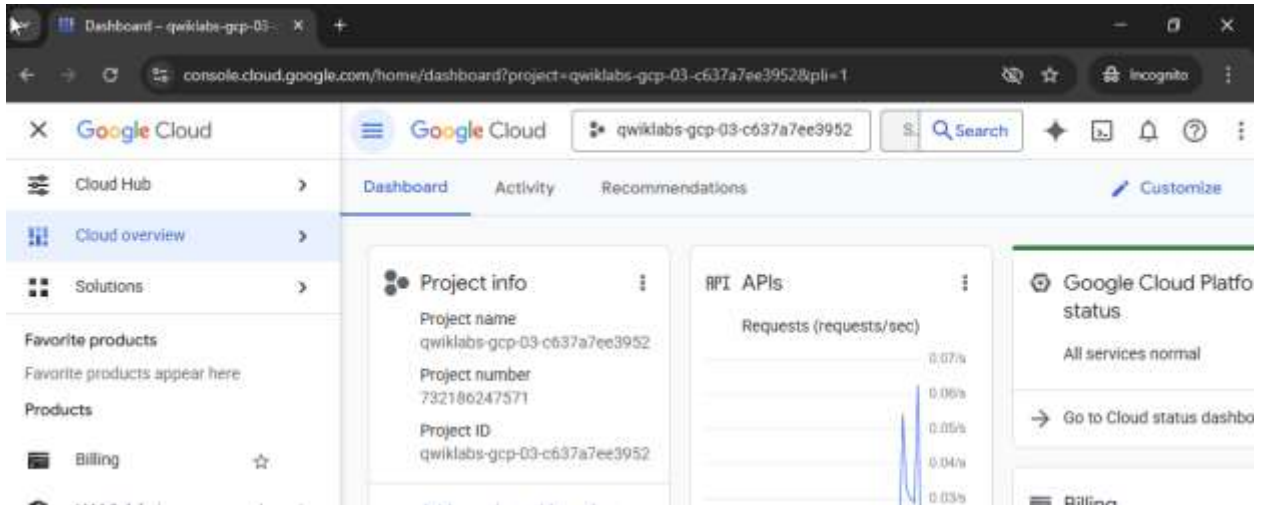
The lab spins up resources, and then opens another tab that shows the **Sign in** page.

3. If necessary, copy the **Username** below and paste it into the **Sign in** dialog. You can also find the **Username** in the **Lab Details** panel.
4. Click Next.
5. Copy the Password below and paste it into the Welcome dialog. You can also find the **Password** in the **Lab Details** panel.
6. Click Next.
7. Click through the subsequent pages:
 - Accept the terms and conditions.



- Do not add recovery options or two-factor authentication (because this is a temporary account).
- Do not sign up for free trials.

After a few moments, the Google Cloud console opens in this tab.



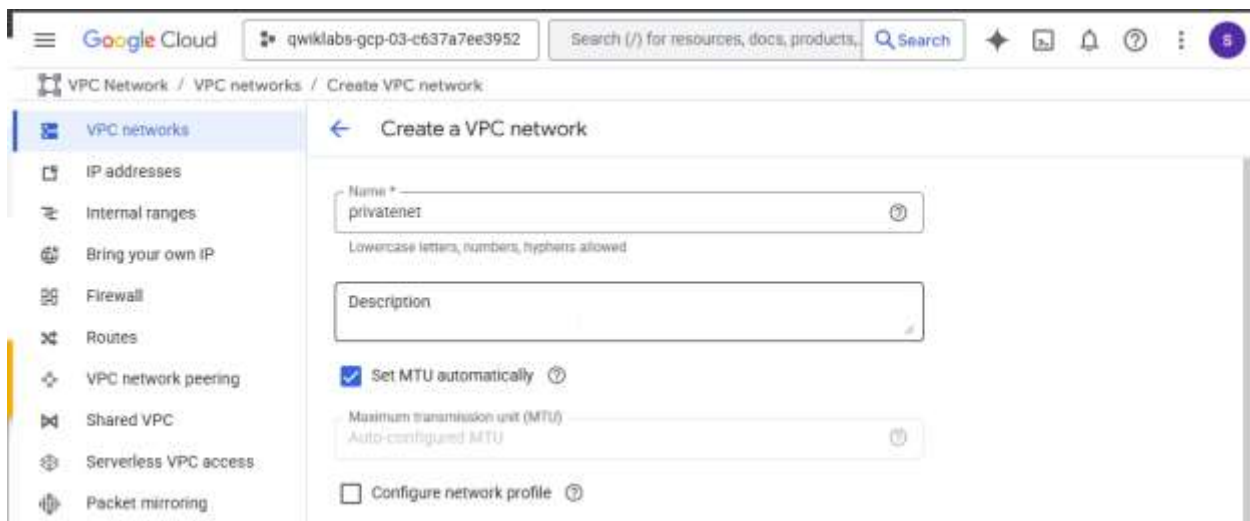
Task 1. Create the VM instance

Create a VPC network with some firewall rules and a VM instance that has no external IP address, and connect to the instance using an IAP tunnel.

Create a VPC network and firewall rules

First, create a VPC network for the VM instance and a firewall rule to allow SSH access.

1. In the Google Cloud console, in the **Navigation menu** (≡), click **VPC network** > **VPC networks**.
2. Click **Create VPC Network**.
3. For **Name**, type **privatenet**.



4. For **Subnet creation mode**, click **Custom**.
5. In **New Subnet** specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Name	privatenet-us
Region	REGION
IPv4 range	10.130.0.0/20

***Note:** Don't enable **Private Google access** yet!*

6. Click **Done**.

Google Cloud

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Search (/) for resources, docs, products.

Search

VPC Network / VPC networks / Create VPC network

VPC networks

IP addresses

Internal ranges

Bring your own IP

Firewall

Routes

VPC network peering

Shared VPC

Serverless VPC access

Packet mirroring

VPC Flow Logs

Create a VPC network

Maximum transmission unit (MTU)
Automatic (Recommended MTU)

Configure network profile

Subnet creation mode

Custom

Automatic

Private IPv6 address settings

Configure a ULA internal IPv6 range for this VPC Network

The ULA range is a /48 CIDR from which all private IPv6 subnet ranges will be taken. Google Cloud can allocate one automatically or you can allocate one manually. Allocation is permanent. You cannot deallocate or change the ULA range.

Subnets
Subnets let you create your own private cloud topology within Google Cloud. Click Automatic to create a subnet in each region, or click Custom to manually define the subnets. [Learn more](#)

Edit subnet

Name
privatenet-us
Lowercase letters, numbers, hyphens allowed

Description

Region
europe-west1 (Belgium)

IP stack type

IPv4 (single-stack)

IPv4 and IPv6 (dual-stack)

IPv6 (single-stack)

Primary IPv4 range

Associate with an internal range

Use an internal range to specify the subnet's internal IP address range. The subnetwork can be associated with an entire internal range or only part of the range.

IPv4 range
10.190.0.0/20
E.g. 10.0.0.0/24

Secondary IPv4 ranges

Add a Secondary IPv4 range

Private Google Access

On

Off

Flow logs

On

Off

Hybrid Subnets

On

Off

Done

7. Click **Create** and wait for the network to be created.

☐ Global
Global routing lets you dynamically learn routes to and from all regions with a single VPC or interconnect and Cloud Router

Best path selection mode ⓘ
☒ Legacy (default)
☐ Standard

DNS configuration (optional)

Managed zone ▼ ⓘ

DNS server policy ▼ ⓘ

Create Cancel

[Equivalent command line](#)
[Equivalent REST](#)

8. In the left pane, click **Firewall**.

Google Cloud

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VPC Network / VPC networks

VPC networks

Create VPC network

Networks in current project

Subnets in current project

Looking for inter-VPC network connectivity at scale? Use NCC for inter-VPC network connectivity.

Network Connectivity Center (NCC) delivers inter-VPC connectivity at 10x the scale of legacy VPC Peering, now supporting up to 250 VPC spokes — with full network isolation and centralized connectivity management model.

Eliminate restrictive peering group quotas, simplify operations with spoke-based architecture, and unlock advanced features like spoke-level range filters, inter-VPC NAT, PSA, and PSC transitivity.

Build your next-generation, scalable, and secure network with NCC — the future-ready solution recommended by Google Cloud.

[Go to NCC](#)
[Learn more](#)
[Remind me later](#)

SMTP port 25 disallowed in this project. [Learn more](#)

VPC networks

Manage flow logs

Filter

Enter property name or value

☐	Name ↑	Subnets	MTU ⓘ	Mode	IPv6
☐	default	24	1460	Auto	
☐	privatenet	1	1460	Custom	

Get started with VPC

Virtual Private Cloud overview

Help document

Get an overview of Virtual Private Cloud networks.

Subnets overview

Help document

Get an overview of subnets.

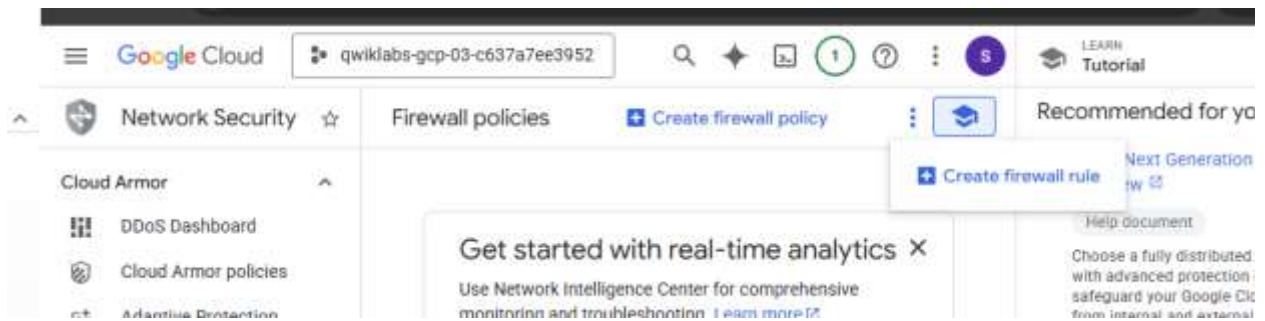
Create and configure Virtual Private Cloud networks and subnets

Help document

Create, update, list, and delete Virtual Private Cloud networks and subnets.

[All VPC documentation](#)

9. Click **Create Firewall Rule**.



10. Specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Name	privatenet-allow-ssh
Network	privatenet
Targets	All instances in the network
Source filter	IPv4 ranges
Source IPv4 ranges	35.235.240.0/20
Protocols and ports	Specified protocols and ports

11. For **tcp**, click the checkbox and specify port **22**.

12. Click **Create**.

Google Cloud

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Search

1

?

...

5

Network Security

Create a firewall rule

Cloud Armor

DDoS Dashboard

Cloud Armor policies

Adaptive Protection

Cloud Armor Service Tier

Cloud IDS

IDS Dashboard

IDS Endpoints

IDS Threats

Cloud NGFW

Dashboard

Firewall policies

Threats

Firewall endpoints

Secure Access Connect

Realms **Preview**

Attachments **Preview**

Cloud NSI

Deployment groups

Endpoint groups

Secure Web Proxy

Web Proxies

SWP Policies

URL Lists

DNS Armor

Threat detection

Common components

Authentication Configur...

Address groups

Security profiles

TLS inspection policies

SSL policies

Firewall rules control incoming and outgoing traffic to an instance. By default, all incoming traffic to your network is blocked. [Learn more](#)

Name *

privatenet-allow-ssh

Lowercase letters, numbers, hyphens allowed

Description

Logs

Turning on firewall logs can generate a large number of logs which can increase costs in Logging. [Learn more](#)

On

Off

Network *

privatenet

Priority *

1000

Compare

Priority can be 0 - 65535

Direction of traffic

Ingress

Egress

Action on match

Allow

Deny

Targets

All instances in the network

Source filter

IPv4 ranges

Source IPv4 ranges *

35.235.240.0/20 X for example, 0.0.0.0/0, 192.168.2.0/24

Second source filter

None

Destination filter

None

Protocols and ports

Allow all

Specified protocols and ports

TCP

Ports

22

E.g. 20, 50-60

UDP

Ports

E.g. all

SCTP

Ports

E.g. 20, 50-60

Other

Protocols

Separate multiple protocols by commas, e.g. all, icmp

Disable rule

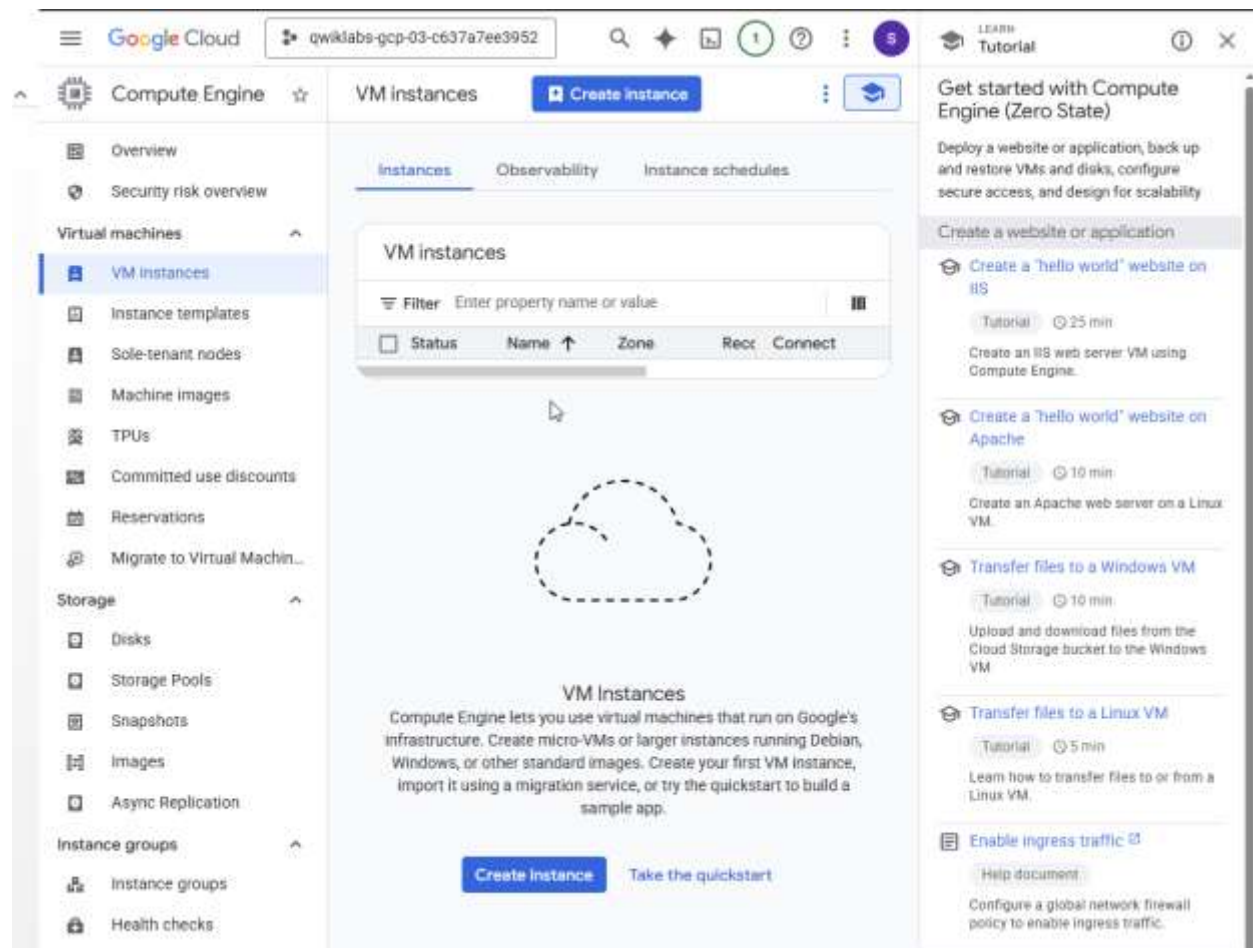
Create

Cancel

Note: In order to connect to your private instance using SSH, you need to open an appropriate port on the firewall. [IAP connections](#) come from a specific set of IP addresses (35.235.240.0/20). Therefore, you can limit the rule to this CIDR range.

Create the VM instance with no public IP address

1. In the Google Cloud console, in the **Navigation menu** () , click **Compute Engine > VM instances**.
2. Click **Create Instance**.



3. On the **Machine configuration** page, specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Name	vm-internal

Region	europa-west1
Zone	europa-west1-b
Series	E2

Machine configuration

e2-standard-2, europe-west1-b

OS and storage

Debian GNU/Linux 12 (bookworm)

Data protection

Snapshot schedules

Networking

1 network interface

Observability

Install Ops Agent

Security

Advanced

Google Cloud

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Search

Create an instance

Create VM from...

Machine configuration

Name *
vm-internal

Region *
europe-west1 (Belgium)

Zone *
europe-west1-b

Region is permanent

Zone is permanent

NEW: Google Axion machine series now in Preview

Try the new N4A series, Google's next generation of Arm-based Axion VMs.

General purpose

Compute optimized

Memory optimized

Storage optimized

GPUs

Machine types for common workloads, optimized for cost and flexibility

	Series	Description	vCPUs	Memory
☐	C4	Consistently high performance	2 - 288	4 - 2,232 GB
☐	C4A	Arm-based consistently high performance	1 - 96	2 - 768 GB
☐	C4D	Consistently high performance	2 - 384	3 - 3,072 GB
☐	N4	Flexible & cost-optimized	2 - 80	4 - 640 GB
☐	N4A	Preview Arm-based flexibility & cost optimization	1 - 64	2 - 512 GB
☐	N4D	Flexible & cost-optimized	2 - 96	4 - 768 GB
☐	C3	Consistently high performance	4 - 192	8 - 1,536 GB
☐	C3D	Consistently high performance	4 - 360	8 - 3,880 GB
☒	E2	Low cost, day-to-day computing	0.25 - 32	1 - 128 GB
☐	N2	Balanced price & performance	2 - 128	2 - 864 GB
☐	N2D	Balanced price & performance	2 - 224	2 - 896 GB
☐	T2A	Scale-out workloads	1 - 48	4 - 192 GB
☐	T2D	Scale-out workloads	1 - 60	4 - 240 GB
☐	N1	Balanced price & performance	0.25 - 96	0.6 - 624 GB

Machine type

Choose a machine type with preset amounts of vCPUs and memory that suit most workloads. Or, you can custom machine for your workload's particular needs. [Learn more](#)

Preset

Custom

e2-standard-2 (2 vCPU, 1 core, 8 GB memory)

vCPU

2 (1 core)

Memory

8 GB

Advanced configurations

Provisioning model

VM provisioning model

Standard

Determines how long your VMs will run. Choice is permanent

VM provisioning model advanced settings

4. Click **OS and storage**.

5. If the **Image** shown is not **Debian GNU/Linux 12 (bookworm)**, click **Change** and select **Debian GNU/Linux 12 (bookworm)**, and then click **Select**.

The screenshot shows the Google Cloud Platform 'Create an instance' page. The 'OS and storage' tab is selected, displaying the 'Debian GNU/Linux 12 (bookworm)' image. The 'Monthly estimate' section shows a total of \$54.81. The 'Networking' tab is visible in the left sidebar.

Item	Monthly estim
2 vCPU + 8 GB memory	\$53
10 GB balanced persistent disk	\$1
Logging	Cost varie
Monitoring	Cost varie
Snapshot schedule	Cost varie
Total	\$54

6. Click **Networking**.
7. In **Network interfaces**, edit the network interface by specifying the following:

Property	Value (type value or select option as specified)
Network	privatenet
Subnetwork	privatenet-us
External IPv4 address	None

Note: The default setting for a VM instance is to have an ephemeral external IP address. This behavior can be changed with a policy constraint at the organization or project level.

8. Click **Done**.
9. Click **Create**, and wait for the VM instance to be created.

Google Cloud

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Search (/) for resources, docs, products, Search

1

Create an instance

Create VM from...

Equivalent code

Machine configuration

e2-standard-2, europe-west1-b

OS and storage

Debian GNU/Linux 12 (bookworm)

Data protection

Snapshot schedules

Networking

1 network interface, privatenet-us (10.130.0.0/20)

Observability

Install Ops Agent

Security

Advanced

Firewall

Select firewall rules to allow specific network traffic from the Internet. Firewall rules will only be applied when an instance is created.

☐ Allow HTTP traffic

☐ Allow HTTPS traffic

☐ Allow Load Balancer Health Checks

Network tags

Network tags

Hostname

Hostname

Set a custom hostname for this instance or leave it default. Choice is permanent.

IP forwarding

☐ Enable

Network performance configuration

Network bandwidth

☐ Enable per VM Tier_1 networking performance

Maximum outbound network bandwidth: 4Gbps

VM to Public IP: 4Gbps

Network interfaces

Network interface is permanent

Edit network interface

Network *

privatenet

Subnetwork *

privatenet-us IPv4 (10.130.0.0/20)

To use IPv6, you need an IPv6 subnet range.

Learn more

Network interface card

IP stack type

☒ IPv4 (single-stack)

☐ IPv4 and IPv6 (dual-stack)

☐ IPv6 (single-stack)

Primary internal IPv4 address

Ephemeral (Automatic)

Alias IP ranges

+ Add IP range

External IPv4 address

None

Done

Add a network interface

Dynamic Network Interfaces

Add a Dynamic Network Interface

Monthly estimate

\$54.81

That's about \$0.08 hourly

Pay for what you use: no upfront costs and per-second billing

Item	Monthly estimate
2 vCPU + 8 GB memory	\$53.81
10 GB balanced persistent disk	\$1.00
Logging	Cost varies
Monitoring	Cost varies
Snapshot schedule	Cost varies
Total	\$54.81

[Compute Engine pricing](#)

[Cloud Operations pricing](#)

Less

Create

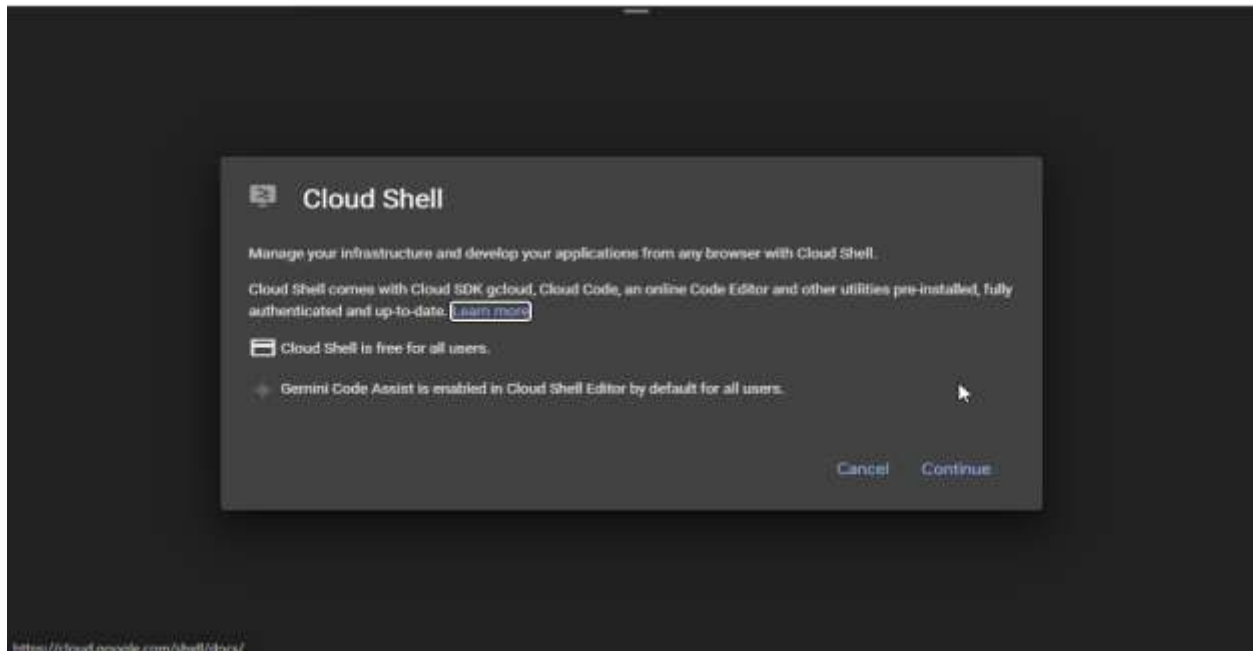
Cancel

Equivalent code

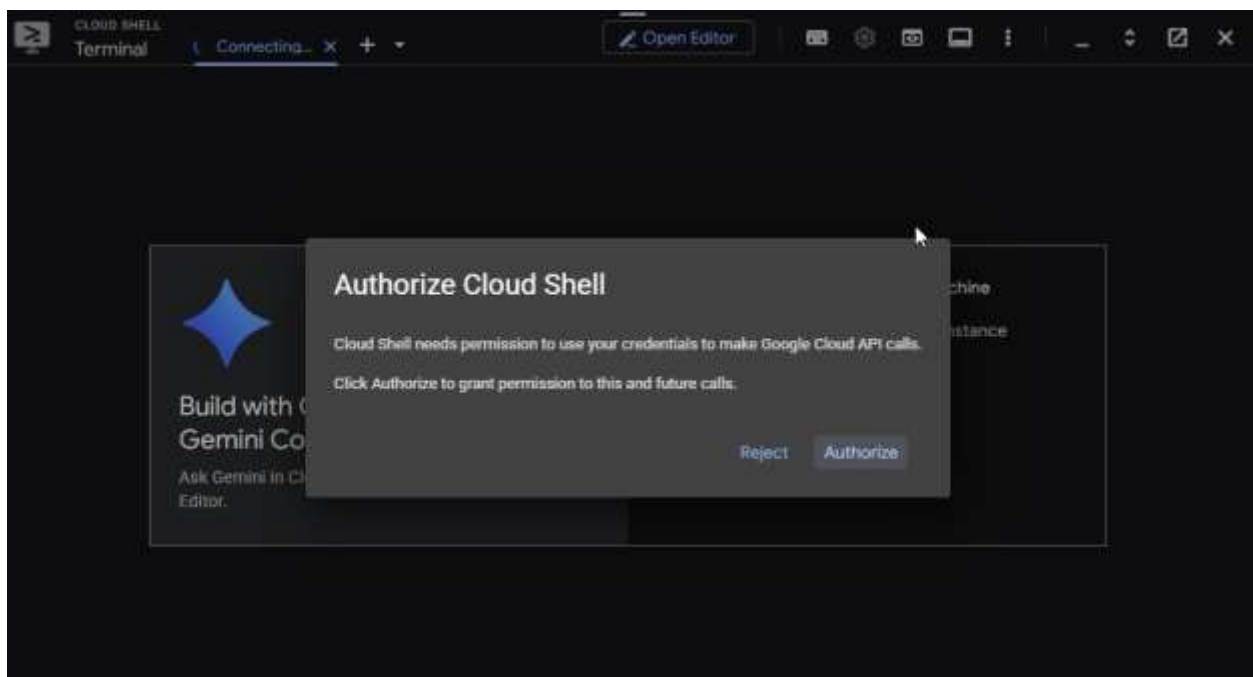
10. On the **VM instances** page, verify that the **External IP** of **vm-internal** is **None**

SSH to **vm-internal** to test the IAP tunnel

1. In the Cloud console, click **Activate Cloud Shell** (▶).
2. If prompted, click **Continue**.



3. If prompted to authorize, click **Authorize**.



5. To connect to **vm-internal**, run the following command:
6. If prompted to continue, type **Y**.
7. When prompted for a passphrase, press **ENTER**.
8. When prompted for the same passphrase, press **ENTER**.

Did the command prompt change to @vm-internal?

True

False

```
You are now logged in as [student-03-552d5185140a@qwiklabs.net].
Your current project is [qwiklabs-gcp-03-c637a7ee3952]. You can change this setting by running:
$ gcloud config set project PROJECT_ID
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$ gcloud compute ssh vm-internal --zone europe-west1-b --tunnel-t
through-lap
WARNING: The private SSH key file for gcloud does not exist.
WARNING: The public SSH key file for gcloud does not exist.
WARNING: You do not have an SSH key for gcloud.
WARNING: SSH keygen will be executed to generate a key.
This tool needs to create the directory [/home/student_03_552d5185140a/.ssh] before being able to generate SSH keys.

Do you want to continue (Y/n)? Y

Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/student_03_552d5185140a/.ssh/google_compute_engine
Your public key has been saved in /home/student_03_552d5185140a/.ssh/google_compute_engine.pub
The key fingerprint is:
SHA256:YBws36dm5C1Ryp03fBD9jQoUa9Tg9V4Wuly3k8aq+as student_03_552d5185140a@cs-995614354964-default
The key's randomart image is:
+----[RSA 3072]-----+
| ..+.. 0. |
| ...+..+... |
| +++.ooo+ . |
| o..oo+oo~o.o |
| . = +$+ o+ * |
| = * * -.- - |
| + = |
| |
| E.. |
+----[SHA256]-----+
WARNING:
To increase the performance of the tunnel, consider installing NumPy. For instructions,
please see https://cloud.google.com/iap/docs/using-tcp-forwarding#increasing_the_tcp_upload_bandwidth

Warning: Permanently added 'compute.8309808766384480571' (ED25519) to the list of known hosts.
Linux vm-internal 6.1.0-42-cloud-amd64 #1 SMP FREEMPT_DYNAMIC Debian 6.1.159-1 (2025-12-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Creating directory '/home/student-03-552d5185140a'.
student-03-552d5185140a@vm-internal:~$
```

9. To test the external connectivity of **vm-internal**, run the following command:

*This should not work because **vm-internal** has no external IP address!*

10. Wait for the **ping** command to complete.
11. To return to your Cloud Shell instance, run the following command:

```
student-03-552d5185140a@vm-internal:~$ ping -c 2 www.google.com
PING www.google.com (172.253.120.104) 56(84) bytes of data.

--- www.google.com ping statistics ---
2 packets transmitted, 0 received, 100% packet loss, time 1014ms

student-03-552d5185140a@vm-internal:~$ exit
logout
Connection to compute.8309808766384480571 closed.
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$
```

Note: When instances do not have external IP addresses, they can only be reached by other instances on the network via a managed VPN gateway or via a Cloud IAP tunnel. Cloud IAP enables context-aware access to VMs via SSH and RDP without bastion hosts.

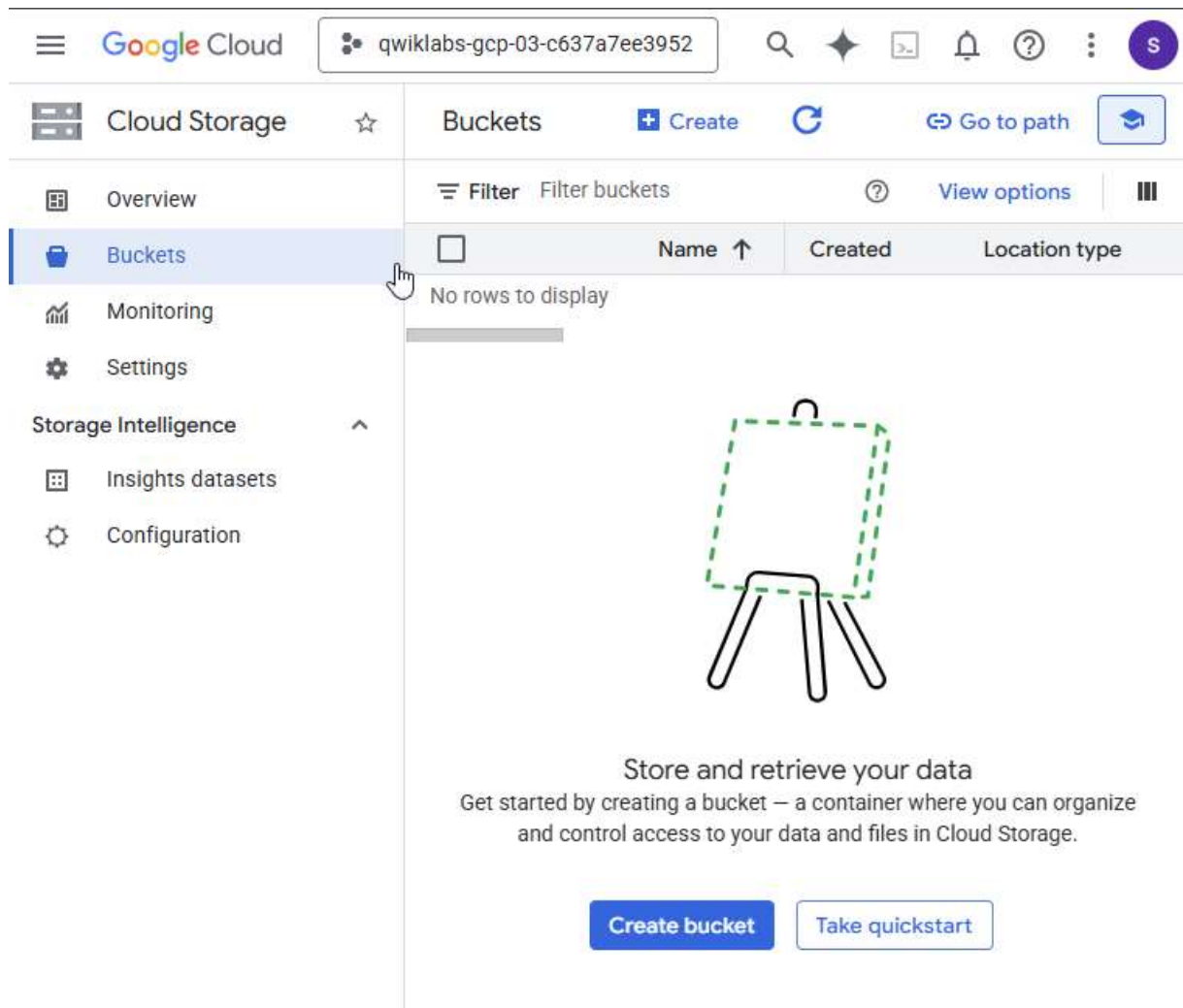
Task 2. Enable Private Google Access

VM instances that have no external IP addresses can use Private Google Access to reach external IP addresses of Google APIs and services. By default, Private Google Access is disabled on a VPC network.

Create a Cloud Storage bucket

Create a Cloud Storage bucket to test access to Google APIs and services.

1. In the Google Cloud console, in the **Navigation menu** () , click **Cloud Storage** > **Buckets**.
2. Click **+Create**.



- Specify the following, and leave the remaining settings as their defaults:

Property	Value (type value or select option as specified)
Name	<i>Enter a globally unique name</i>
Location type	Multi-region

- Click **Create**.

Cloud Storage

Overview

Buckets

Monitoring

Settings

Storage Intelligence

Insights datasets

Configuration

Create a bucket

✓

Get Started

Name: lonebuck

•

Choose where to store your data

This choice defines the geographic placement of your data and affects cost, performance, and availability. Cannot be changed later. [Learn more](#)

•

Only locations allowed by the organization's resource location policy can be selected.

Location type

•

Multi-region

Highest availability across largest area

us (multiple regions in United States)

Add cross-bucket replication via Storage Transfer Service

As data is added or changed, replicate it to another bucket, enabling you to store a copy that follows different bucket settings - e.g., region, storage class, etc. [Learn more](#)

Dual-region

High availability and low latency across 2 regions

Region

Lowest latency within a single region

Continue

•

Choose how to store your data

Default storage class: Standard

Hierarchical namespace: Disabled

Anywhere Cache: Disabled

•

Choose how to control access to objects

Public access prevention: On

Access control: Uniform

•

Choose how to protect object data

Soft delete policy: Default

Object versioning: Disabled

Bucket retention policy: Disabled

Object retention: Disabled

Encryption type: Google-managed

Good to know

•

Location pricing

Storage rates vary depending on the storage class of your data and location of your bucket. [Pricing details](#)

Current configuration: Multi-region / Standard

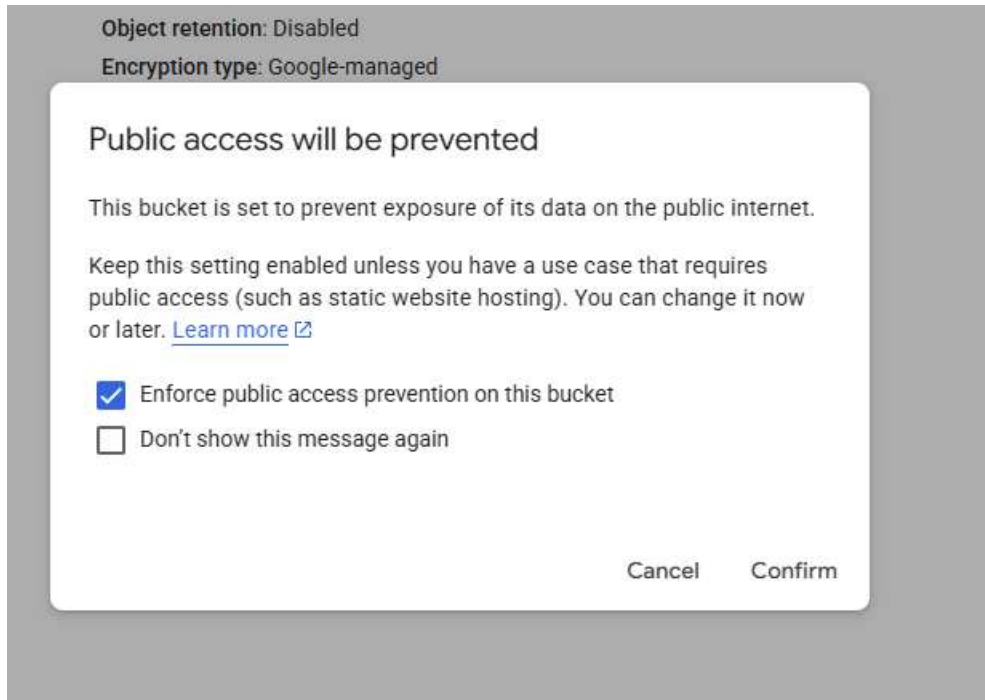
Item	Cost
us (multiple regions in United States)	\$0.026 per GB-month
With default replication	\$0.020 per GB written

[Estimate your monthly cost](#)

Marketplace

CreateCancel

If prompted to enable public access prevention, ensure it is checked and click **Confirm**. Note the name of your storage bucket.



5. Store the name of your bucket in an environment variable:
6. Verify it with echo:

```
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952) $ export MY_BUCKET=lonchuck
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952) $ echo $MY_BUCKET
lonchuck
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952) $
```

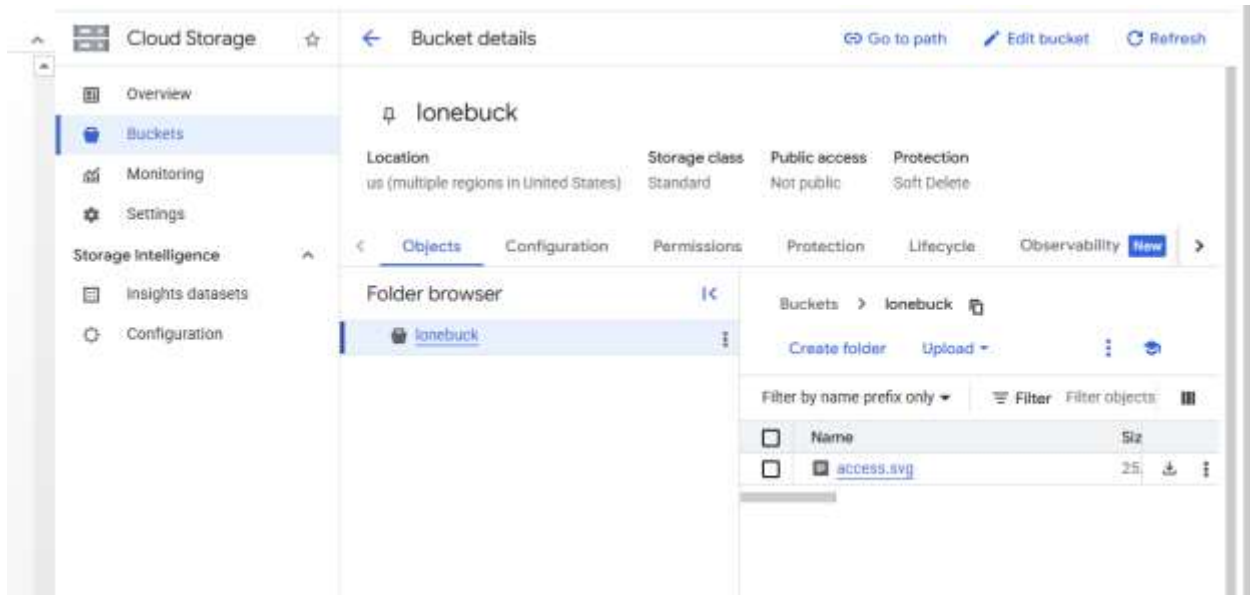
Copy an image file into your bucket

Copy an image from a public Cloud Storage bucket to your own bucket.

1. In Cloud Shell, run the following command:

```
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952) $ gcloud storage cp gs://cloud-training/gcpnet/private/access.svg
gs://$MY_BUCKET
Copying gs://cloud-training/gcpnet/private/access.svg to gs://lonchuck/access.svg
Completed files 1/1 | 24.8kiB/24.8kiB
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952) $
```

2. In the Cloud console, click your bucket name to verify that the image was copied. You can click on the name of the image in the Cloud console to view an example of how Private Google Access is implemented.



Access the image from your VM instance

Currently, which of your VM instances can access the image from your bucket?



Cloud Shell



vm-internal

1. In Cloud Shell, to try to copy the image from your bucket, run the following command:

This should work because Cloud Shell has an external IP address!

```
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$ gcloud storage cp gs://$MY_BUCKET/*.svg .
Copying gs://lonebuck/access.svg to file:///./access.svg
Completed files 1/1 | 24.8kiB/24.8kiB
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$
```

I

2. To connect to vm-internal, run the following command:
3. If prompted, type **Y** to continue.


```
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$ gcloud compute ssh vm-internal --zone europe-west1-b --tunnel-through-iap
WARNING:

To increase the performance of the tunnel, consider installing NumPy. For instructions,
please see https://cloud.google.com/iap/docs/using-tcp-forwarding#increasing_the_tcp_upload_bandwidth

Linux vm-internal 6.1.0-42-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.159-1 (2025-12-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jan 17 14:51:39 2026 from 35.235.247.33
student-03-552d5185140a@vm-internal:~$
```

4. Store the name of your bucket in an environment variable:
5. Verify it with echo:
6. Try to copy the image to **vm-internal**, run the following command:

This should not work: vm-internal can only send traffic within the VPC network because Private Google Access is disabled (by default).

7. Press Ctrl+C to stop the request.

```
student-03-552d5185140a@vm-internal:~$ export MY_BUCKET=lonebuck
student-03-552d5185140a@vm-internal:~$ echo $MY_BUCKET
lonebuck
student-03-552d5185140a@vm-internal:~$ gcloud storage cp gs://$MY_BUCKET/*.svg .
^C

Command killed by keyboard interrupt

student-03-552d5185140a@vm-internal:~$
```

Enable Private Google Access

Private Google Access is enabled at the subnet level. When it is enabled, instances in the subnet that only have private IP addresses can send traffic to Google APIs and services through the default route (0.0.0.0/0) with a next hop to the default internet gateway.

1. In the Cloud console, in the **Navigation menu** (☰), click **VPC network > VPC networks**.
2. Click **privatenet** to open the network.
3. Click **Subnets**, and then click **privatenet-us**.
4. Click **Edit**.
5. For **Private Google access**, select **On**.
6. Click **Save**.

Note: Enabling Private Google Access is as simple as selecting **On** within the subnet!

VPC Network / Subnetwork: privatenet-us

VPC networks

IP addresses

Internal ranges

Bring your own IP

Firewall

Routes

VPC network peering

Shared VPC

Serverless VPC access

Packet mirroring

VPC Flow Logs

Subnet details

Edit

Delete

privatenet-us

VPC Network
privatenet

Region
europe-west1

IP stack type
☒ IPv4 (single-stack)
☐ IPv4 and IPv6 (dual-stack) ?

Primary IPv4 range

Subnetwork IP ranges must be unique and non-overlapping within a VPC network and peered VPC network. The following ranges are currently being used in other regions: None

Primary IPv4 range *
10.130.0.0/20

Secondary IPv4 ranges ?

Add a Secondary IPv4 range

Gateway
10.130.0.1

Private Google Access
☒ On
☐ Off

Hybrid Subnets ?
☐ On
☒ Off

Tags

Save Cancel

- Run the following command, in **Cloud Shell** for **vm-internal**, to try to copy the image to **vm-internal**.

This should work because vm-internal's subnet has Private Google Access enabled!

8. To return to your Cloud Shell instance, run the following command:

```
student-03-552d5185140a@vm-internal:~$ gcloud storage cp gs://$MY_BUCKET/*.svg .
Copying gs://lonebuck/access.svg to file:///./access.svg
Completed files 1/1 | 24.8kiB/24.8kiB
student-03-552d5185140a@vm-internal:~$ exit
logout
Connection to compute.8309808766384480571 closed.
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952) $
```

Task 3. Configure a Cloud NAT gateway

Although vm-internal can now access certain Google APIs and services without an external IP address, the instance cannot access the internet for updates and patches. Configure a Cloud NAT gateway, which allows vm-internal to reach the internet.

Try to update the VM instances

1. In Cloud Shell, to try to re-synchronize the package index, run the following:

```
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952) $ sudo apt-get update
*****
You are running apt-get inside of Cloud Shell. Note that your Cloud Shell
machine is ephemeral and no system-wide change will persist beyond session end.

To suppress this warning, create an empty ~/.cloudshell/no-apt-get-warning file.
The command will automatically proceed in 5 seconds or on any key.

Visit https://cloud.google.com/shell/help for more information.
*****
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 https://packages.cloud.google.com/apt gcsfuse-noble InRelease [1,227 B]
Get:5 https://apt.postgresql.org/pub/repos/apt noble-pgdg InRelease [107 kB]
Get:6 https://packages.cloud.google.com/apt cloud-sdk InRelease [1,620 B]
Get:7 https://ppa.launchpadcontent.net/dotnet/backports/ubuntu noble InRelease [24.1 kB]
Get:8 https://cli.github.com/packages stable InRelease [3,917 B]
Get:9 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [2,142 kB]
Get:10 http://archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [3,077 kB]
Get:11 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,959 kB]
Get:12 https://packages.cloud.google.com/apt cloud-sdk/main amd64 Packages [4,414 kB]
Get:13 https://apt.postgresql.org/pub/repos/apt noble-pgdg/main amd64 Packages [579 kB]
Get:14 https://packages.cloud.google.com/apt cloud-sdk/main all Packages [1,911 kB]
Get:15 https://ppa.launchpadcontent.net/dotnet/backports/ubuntu noble/main amd64 Packages [6,476 B]
Get:16 https://cli.github.com/packages stable/main amd64 Packages [354 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:18 https://download.docker.com/linux/ubuntu noble InRelease [48.5 kB]
Get:19 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0 InRelease [3,005 B]
Get:20 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1,191 kB]
Get:21 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2,919 kB]
Get:22 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1,769 kB]
Get:23 https://download.docker.com/linux/ubuntu noble/stable amd64 Packages [50.7 kB]
Get:24 https://repo.mongodb.org/apt/ubuntu noble/mongodb-org/8.0/multiverse amd64 Packages [58.3 kB]
Fetched 20.6 MB in 6s (3,518 kB/s)
Reading package lists... Done
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952) $
```

2. To connect to vm-internal, run the following command:
3. If prompted, type Y to continue.

```

student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$ gcloud compute ssh vm-internal --zone europe-west1-b --tunnel-through-lap
WARNING:

To increase the performance of the tunnel, consider installing NumPy. For instructions,
please see https://cloud.google.com/iap/docs/using-tcp-forwarding#increasing_the_tcp_upload_bandwidth

Linux vm-internal 6.1.0-42-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.159-1 (2025-12-30) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jan 17 15:07:33 2026 from 35.235.247.33
student-03-552d5185140a@vm-internal:~$

```

4. To try to re-synchronize the package index of vm-internal, run the following command:

This should only work for Google Cloud packages because vm-internal only has access to Google APIs and services!

```

student-03-552d5185140a@vm-internal:~$ sudo apt-get update
Reading package lists... Done
E: Could not get lock /var/lib/apt/lists/lock. It is held by process 1357 (apt-get)
N: Be aware that removing the lock file is not a solution and may break your system.
E: Unable to lock directory /var/lib/apt/lists/
student-03-552d5185140a@vm-internal:~$

```

Configure a Cloud NAT gateway

Cloud NAT is a regional resource. You can configure it to allow traffic from all ranges of all subnets in a region, from specific subnets in the region only, or from specific primary and secondary CIDR ranges only.

1. On the Google Cloud console title bar, type **Network services** in the **Search** field, then click **Network services** in the **Products & Page** section.

network services



 Search

Top results



Network Services

Product · Network management tools



Load balancing

Product page · Network Services



Cloud DNS

Product page · Network Services

Products & pages

[Show more](#)



Network Services

Product · Network management tools



Load balancing

Product page · Network Services



Cloud DNS

Product page · Network Services

Documentation & tutorials

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Network Service Tiers overview

Documentation · Use Premium Tier to optimize for performance, and use...



Networking technologies

Documentation · Documentation and resources for Google Cloud products that...



Migrate a network load balancer from target pool to a regional backend service

Tutorial

Marketplace

[Show more](#)



MRVN

Marketplace · MRVN is a cloud-native platform that simplifies IT ops across mult...



Network Services API

Marketplace



Cloud Virtual Network

Marketplace · Managed networking functionality for your Google Cloud Platform...

Showing resource results for  qwiklabs-gcp-03-c637a7ee3952 only [?](#)

2. On the **Network service** page, click **Pin** next to Network services.
3. Click **Cloud NAT**.
4. Click **Get started** to configure a NAT gateway.
5. Specify the following:

Property	Value (type value or select option as specified)
Gateway name	nat-config
Network	privatenet
Region	

Network Services ☆

Load balancing

Cloud DNS

Cloud CDN

Cloud NAT

Cloud Service Mesh (Traff...

Service Directory

Cloud Domains

Private Service Connect

SSL policies

Service Extensions

← Create Cloud NAT gateway Learn

Cloud NAT lets your VMs and container pods create outbound connections to the internet or to other Virtual Private Cloud (VPC) networks.

Cloud NAT uses Cloud NAT gateway to manage those connections. Cloud NAT gateway is region and VPC network specific. If you have VM instances in multiple regions, you'll need to create a Cloud NAT gateway for each region. [Learn more](#)

Gateway name *

nat-config

Lowercase letters, numbers, hyphens allowed

NAT type

☒ Public

☐ Private

Select Cloud Router ?

Network *

privatenet

Region *

europa-west1 (Belgium)

6. For **Cloud Router**, select **Create new router**.

7. For **Name**, type **nat-router**

8. Click **Create**.

wiklab

Create a router ✕

Google Cloud Router dynamically exchanges routes between your Virtual Private Cloud (VPC) and on-premises networks by using Border Gateway Protocol (BGP)

Name *

nat-router ?

Lowercase letters, numbers, hyphens allowed

Description

Network

privatenet ?

Region

europa-west1 (Belgium)

BGP peer keepalive interval

seconds ?

BGP identifier ?

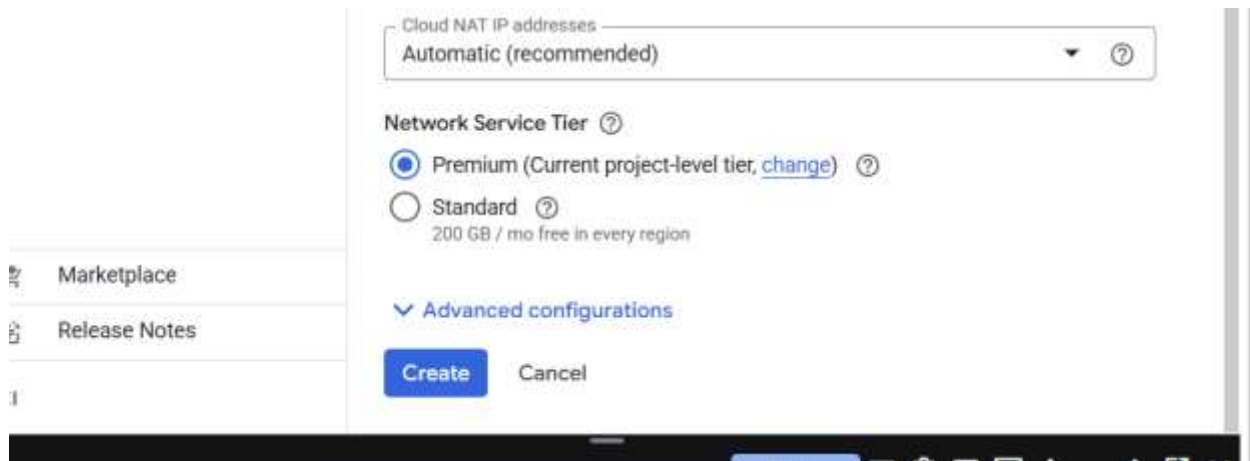
E.g. 169.254.16.16/30. If not specified, Google will automatically-assign an IPv4 range.

Create

Cancel

*Note: The **NAT mapping** section allows you to choose the subnets to map to the NAT gateway. You can also manually assign static IP addresses that should be used when performing NAT. Do not change the NAT mapping configuration in this lab.*

9. Click **Create**.



10. Wait for the gateway's status to change to **Running**.

Filter Enter property name or value ? ⋮					
Network	Region	NAT type	Cloud router	Status	
privatenet	europe-west1	Public	nat-router	✓ Running	⋮

Verify the Cloud NAT gateway

It may take up to 3 minutes for the NAT configuration to propagate to the VM, so wait at least a minute before trying to access the internet again.

1. In Cloud Shell for `vm-internal`, to try to re-synchronize the package index of `vm-internal`, run the following command:

*This should work because **vm-internal** is using the NAT gateway!*

2. To return to your Cloud Shell instance, run the following command:

```
student-03-552d5185140a@vm-internal:~$ sudo apt-get update
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [30 B]
Get:2 file:/etc/apt/mirrors/debian-security.list Mirrorlist [39 B]
Hit:3 https://deb.debian.org/debian bookworm InRelease
Hit:7 https://packages.cloud.google.com/apt google-compute-engine-bookworm-stable InRelease
Hit:4 https://deb.debian.org/debian bookworm-updates InRelease
Hit:5 https://deb.debian.org/debian bookworm-backports InRelease
Hit:6 https://deb.debian.org/debian-security bookworm-security InRelease
Hit:8 https://packages.cloud.google.com/apt cloud-sdk-bookworm InRelease
Hit:9 https://packages.cloud.google.com/apt google-cloud-ops-agent-bookworm-2 InRelease
Reading package lists... Done
student-03-552d5185140a@vm-internal:~$ exit
logout
Connection to compute.8309808766384480571 closed.
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952) $
```

Note: The Cloud NAT gateway implements outbound NAT, but not inbound NAT. In other words, hosts outside of your VPC network can only respond to connections initiated by your instances; they cannot initiate their own, new connections to your instances via NAT.

Task 4. Configure and view logs with Cloud NAT Logging

Cloud NAT logging allows you to log NAT connections and errors. When Cloud NAT logging is enabled, one log entry can be generated for each of the following scenarios:

- When a network connection using NAT is created.
- When a packet is dropped because no port was available for NAT.

You can opt to log both kinds of events, or just one or the other. Created logs are sent to Cloud Logging.

Enabling logging

If logging is enabled, all collected logs are sent to Cloud Logging by default. You can filter these so that only certain logs are sent.

You can also specify these values when you create a NAT gateway or by editing one after it has been created. The following directions show how to enable logging for an existing NAT gateway.

1. In the Google Cloud console, in the **Navigation menu** () , click **Network services** > **Cloud NAT**.
2. Click on the nat-config gateway and then click **Edit**.
3. Click the **Advanced configurations** dropdown to open that section.
4. For **Logging**, select **Translation and errors** and then click **Save**.

- Load balancing
- Cloud DNS
- Cloud CDN
- Cloud NAT**
- Cloud Service Mesh (Traff...
- Service Directory
- Cloud Domains
- Private Service Connect
- SSL policies
- Service Extensions

- Marketplace
- Release Notes

← Edit Cloud NAT gateway

Automatic (recommended) ▼ ?

Network Service Tier ?

- ☒ Premium (Current project-level tier, [change](#)) ?
- ☐ Standard ?
200 GB / mo free in every region

Advanced configurations

Logging ?

Log Cloud NAT events

- ☐ No logging
- ☒ Translation and errors
- ☐ Translation only
- ☐ Errors only

Port allocation

Cloud NAT gateways allocate source ports to Compute Engine virtual machine (VM) instance and Google Kubernetes Engine (GKE) nodes that use the gateways. Enabling dynamic port allocation will allow you to use custom maximum ports per VM instance.

☐ Enable Dynamic Port Allocation ?Min ports per VM
64 (Default) ?

Max ports per VM ▼ ?

☐ Enable Endpoint-Independent Mapping ?

Timeouts for protocol connections

UDP
30 (Default) seconds ?TCP established
1200 (Default) seconds ?TCP transitory
30 (Default) seconds ?TCP time wait
120 (Default) seconds ?ICMP
30 (Default) seconds ?

^ Hide advanced configurations

Save Cancel

NAT logging in Cloud Logging

Now that you have set up Cloud NAT logging for the nat-config gateway, let's find out where we can view our logs.

1. Click on nat-config to expose its details. Then click on the **View in Logs Explorer**.

The screenshot shows the Google Cloud Platform console interface. On the left is a sidebar with 'Network Services' selected, containing links to Load balancing, Cloud DNS, Cloud CDN, Cloud NAT (highlighted), Cloud Service Mesh (Traffic...), Service Directory, Cloud Domains, Private Service Connect, SSL policies, and Service Extensions. The main content area is titled 'Cloud NAT ga...' and includes an 'Edit' button and an 'Investigate' button. Below the title, a green checkmark indicates the 'nat-config' gateway is active. A table shows 'Status' as 'Running' and 'Logs' with a link to 'View in Logs Explorer'. There are two tabs: 'Details' (selected) and 'Monitoring'. Under 'Details', there are three sections: 'NAT type' with 'Type' as 'Public'; 'Cloud Router' with 'Region' as 'europe-west1', 'VPC network' as 'privatenet', and 'Cloud Router' as 'nat-router'; and 'Cloud NAT mapping' with 'High availability' as 'Yes', 'Source endpoint type' as 'VM instances, GKE nodes, Serverless', and 'Source IPv4 subnets' as 'All subnets' primary and secondary IP ranges'.

Network Services	
Load balancing	
Cloud DNS	
Cloud CDN	
Cloud NAT	
Cloud Service Mesh (Traffic...)	
Service Directory	
Cloud Domains	
Private Service Connect	
SSL policies	
Service Extensions	

← Cloud NAT ga... Edit Investigate

✓ nat-config

Status	Running
Logs	View in Logs Explorer

Details Monitoring

NAT type

Type	Public
------	--------

Cloud Router

Region	europe-west1
VPC network	privatenet
Cloud Router	nat-router

Cloud NAT mapping

High availability	Yes
Source endpoint type	VM instances, GKE nodes, Serverless
Source IPv4 subnets	All subnets' primary and secondary IP ranges

2. This will open a new tab with **Logs Explorer**.

You will see that there aren't any logs yet—that's because we just enabled this feature for the gateway.

Note: Keep this tab open and return to your other Google Cloud console tab.

Generating logs

As a reminder, Cloud NAT logs are generated for the following sequences:

- When a network connection using NAT is created.
- When a packet is dropped because no port was available for NAT.

Let's connect the host to the internal VM again to see if any logs are generated.

1. In **Cloud Shell** for **vm-internal**, to try to re-synchronize the package index of **vm-internal**, run the following command:
2. If prompted, type **Y** to continue.
3. Try to re-synchronize the package index of **vm-internal** by running the following:
4. To return to your Cloud Shell instance, run the following command:

```

student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$ gcloud compute ssh vm-internal --zone europe-west1-b --tunnel-through-iap
WARNING:

To increase the performance of the tunnel, consider installing NumPy. For instructions,
please see https://cloud.google.com/iap/docs/using-tcp-forwarding#increasing_the_tcp_upload_bandwidth

Linux vm-internal 6.1.0-42-cloud-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.159-1 (2025-12-30) x86_64

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individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Sat Jan 17 15:17:47 2026 from 35.235.247.32
student-03-552d5185140a@vm-internal:~$ sudo apt-get update
Get:1 file:/etc/apt/mirrors/debian.list Mirrorlist [30 B]
Get:3 file:/etc/apt/mirrors/debian-security.list Mirrorlist [39 B]
Hit:2 https://deb.debian.org/debian bookworm InRelease
Hit:7 https://packages.cloud.google.com/apt google-compute-engine-bookworm-stable InRelease
Hit:4 https://deb.debian.org/debian bookworm-updates InRelease
Hit:5 https://deb.debian.org/debian bookworm-backports InRelease
Hit:6 https://deb.debian.org/debian-security bookworm-security InRelease
Hit:8 https://packages.cloud.google.com/apt cloud-sdk-bookworm InRelease
Hit:9 https://packages.cloud.google.com/apt google-cloud-ops-agent-bookworm-2 InRelease
Reading package lists... Done
student-03-552d5185140a@vm-internal:~$ eict
-bash: eict: command not found
student-03-552d5185140a@vm-internal:~$ exit
logout
Connection to compute.8309808766384480571 closed.
student_03_552d5185140a@cloudshell:~ (qwiklabs-gcp-03-c637a7ee3952)$

```

Let's see if opening up this connection revealed anything new in our logs.

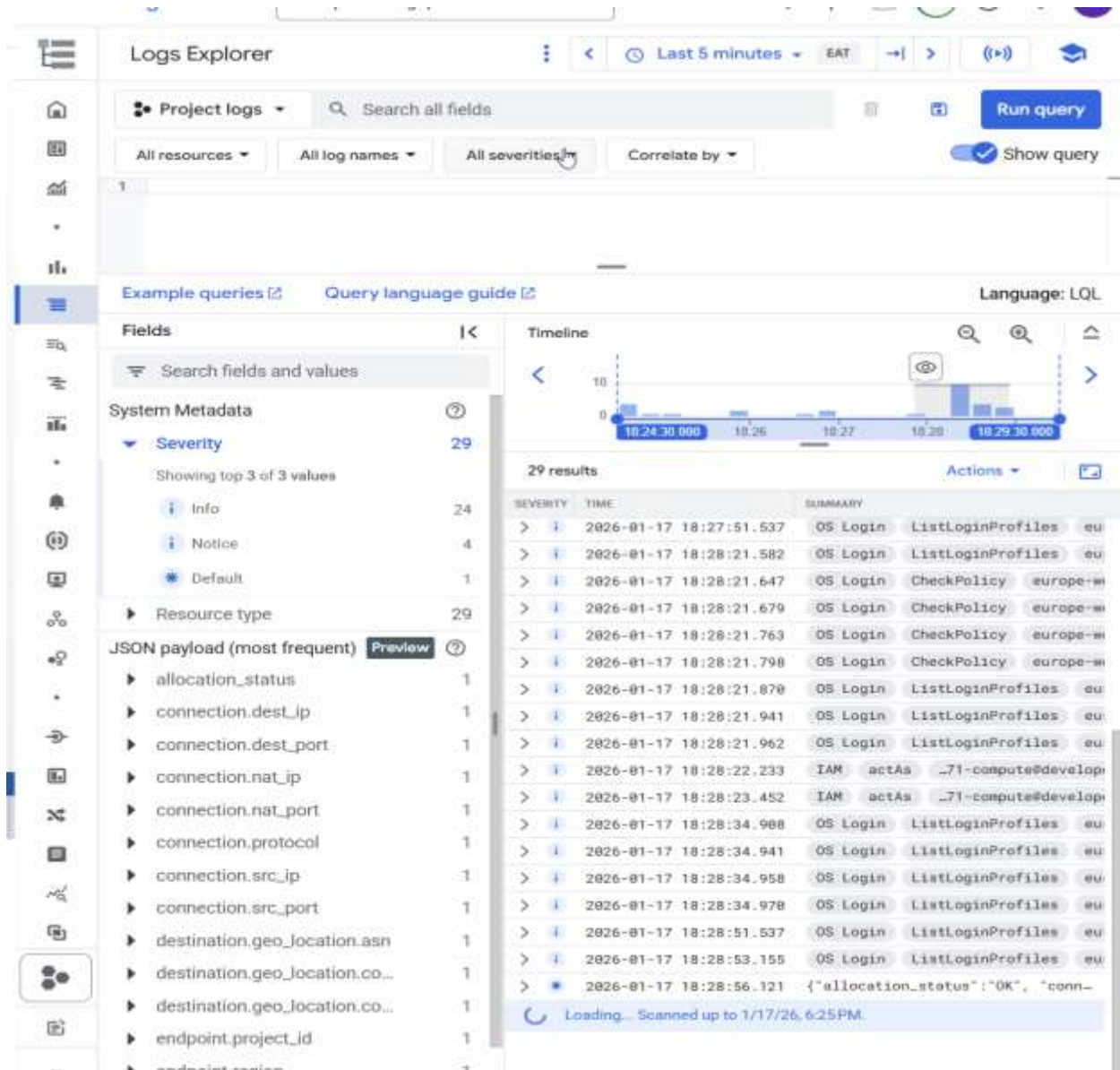
Viewing Logs

- Return to your Logs Explorer tab, and in the navigation menu, click **Logs Explorer**.

You should see two new logs that were generated after connecting to the internal VM.

Note: You may need to wait for a few minutes. If you are still unable to see the logs, repeat step 1 to step 4, from the **Generating logs** section, and then refresh the logging page.

As we see, the logs give us details on the VPC network we connected to and the connection method we used. Feel free to expand different labels and details.



Task 5. Review

I created **vm-internal**, an instance with no external IP address, and connected to it securely using an IAP tunnel. Then I enabled Private Google Access, configured a NAT gateway, and verified that **vm-internal** can access Google APIs and services and other public IP addresses.

VM instances without external IP addresses are isolated from external networks. Using Cloud NAT, these instances can access the internet for updates and patches, and in some cases, for bootstrapping. As a managed service, Cloud NAT provides high availability without user management and intervention. IAP uses your existing project roles and permissions when you connect to VM instances. By default, instance owners are the only users that have the **IAP Secured Tunnel User** role.

